Optics, Camera Models & Location Tools

Lecture 3

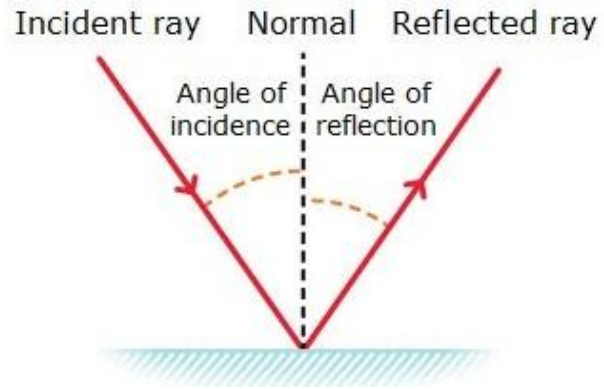
Light

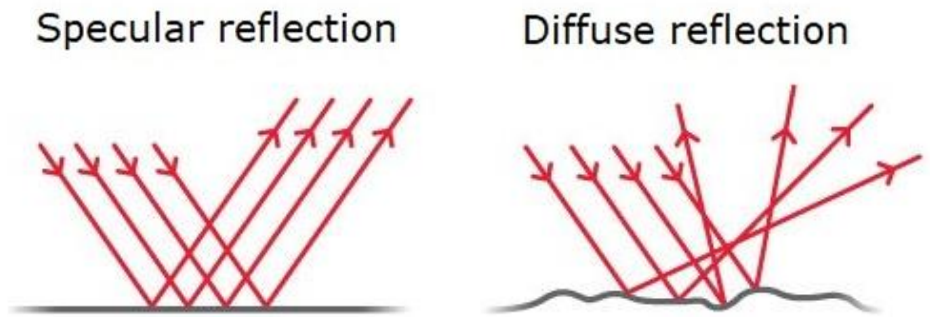
- **Light** is **electromagnetic radiation** visible to the human eye, with **wavelengths** ranging from approximately 400 nm (violet) to 700 nm (red).
- A triangular prism disperses a beam of **white light** into its component wavelengths. The colors are separated according to their wavelengths, with red having the longest wavelength and violet having the shortest.
- In machine vision, light allows us to **capture images** by reflecting from or transmitting through objects.



Optics

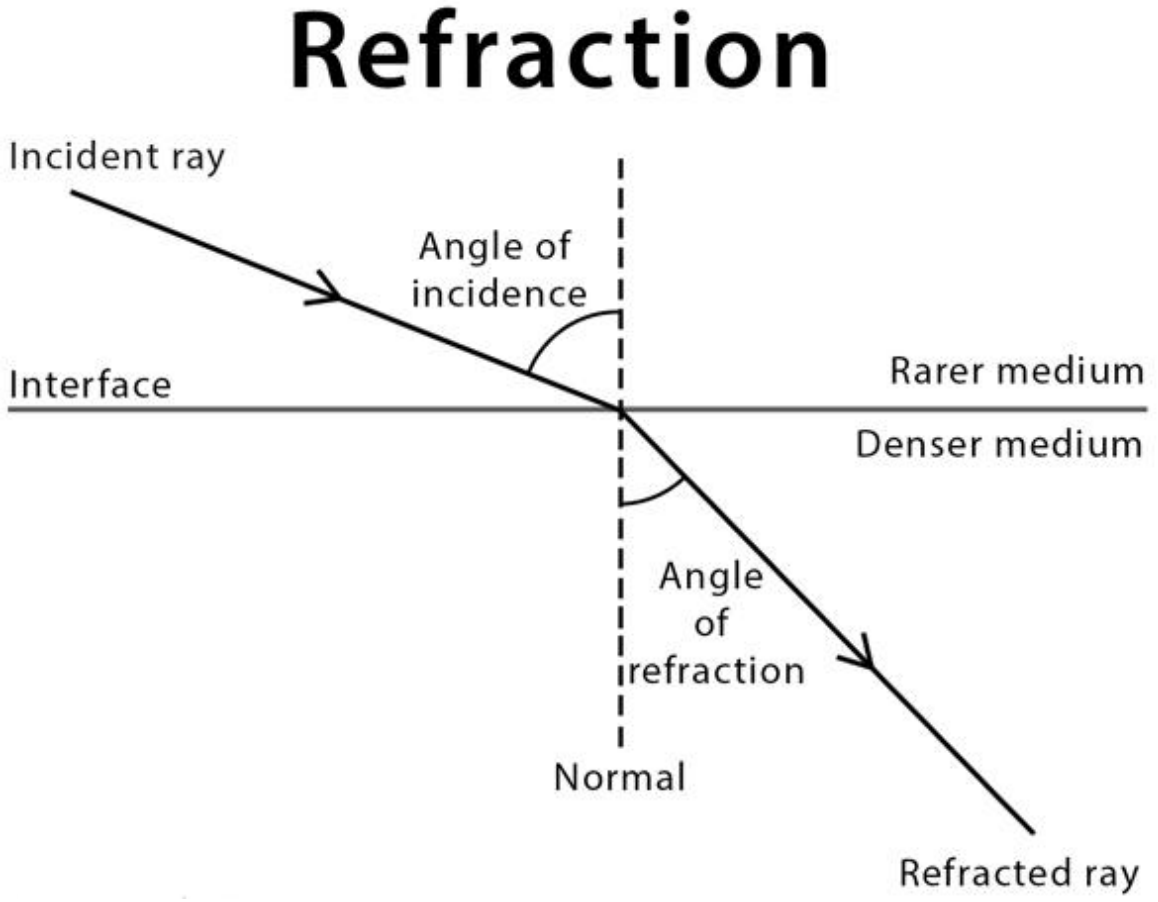
- **Reflection**: When light strikes an object, some of the light is reflected from the surface. The nature of the reflection depends on the surface properties.
  - **Specular Reflection**: Occurs on smooth surfaces, where light is reflected in a well-defined direction, with the angle of reflection equal to the angle of incidence.
  - **Diffuse Reflection**: Occurs on rough surfaces, where light is reflected in many different directions due to variations in the surface orientation.





Optics

- The **remaining light** may be **absorbed by** or **transmitted through** the object, depending on the material.
- **Absorption**: Objects absorb certain wavelengths of light, which contributes to their color appearance.
- **Transmission**: Transparent or translucent materials allow light to pass through them.
- **Refraction**: The bending of light as it passes through materials of different densities (e.g., from air to glass).



Real-World vs. Camera Image

- A **camera** is a device that captures light from the 3D real-world and records it as an image.
- Real-world is described in physical units, while the image is described in pixels.
- How do we relate them so that computers can obtain the real-world information from images?

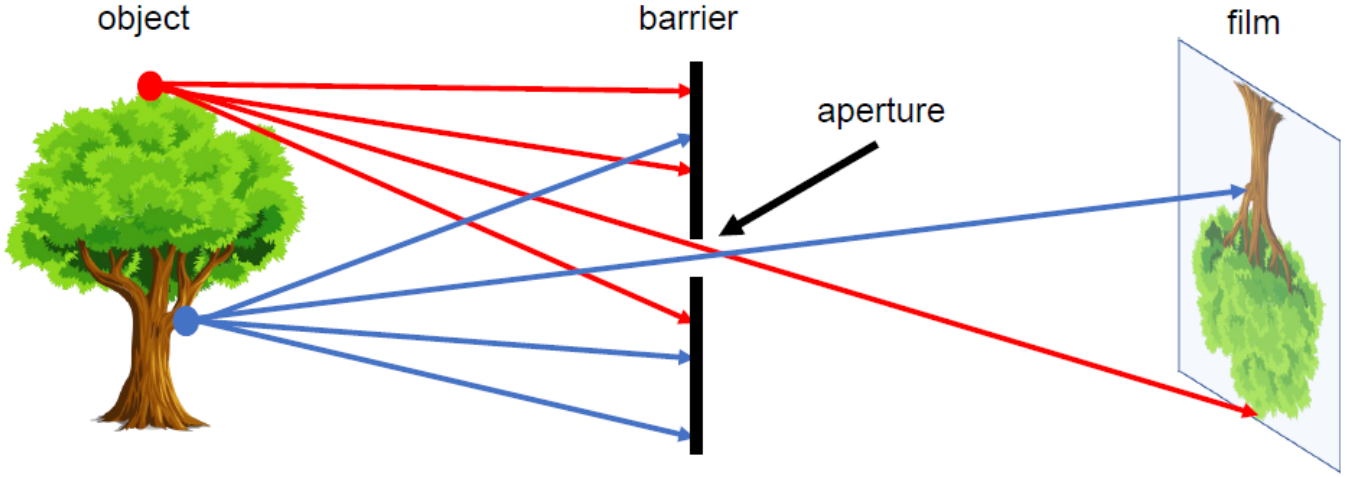
Camera Models

- **Camera models** mathematically describe how light from the 3D world gets transformed into an image.
- Without a camera model, we can’t relate image pixels to the real-world geometry.
- Two models:
  - Pinhole camera model
  - Camera model with lens

Pinhole Camera Model

- **Pinhole Camera Model**: A simplified representation in which light passes through a small aperture (pinhole) and forms an inverted image on the image plane (sensor plane).
- It helps us understand the **geometric relationships involved in image formation**.

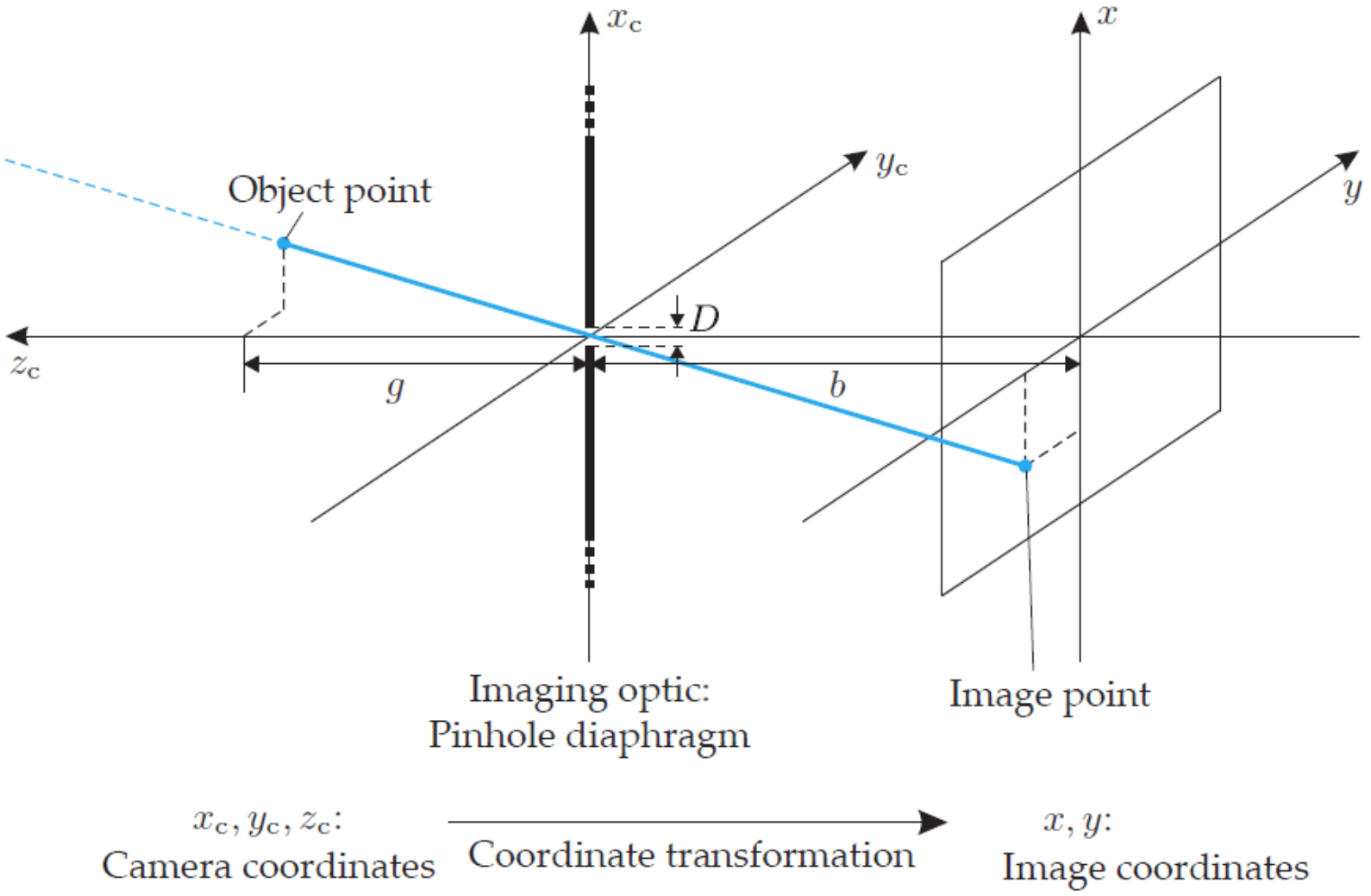




Pinhole Camera Model

- In this model, the optical mapping of a 3D object point to a 2D image point can be mathematically described using **central projection** (projection of 3D points onto a 2D image plane through a single projection center):

$$\begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix} = -\frac{b}{z_c} \begin{pmatrix} x_c \\ y_c \end{pmatrix}$$



Pinhole Camera Model - Transformation from the Image Plane to the Digital Image - Intrinsic Parameters

- Points in the **image plane** and the **digital image** generally use different coordinate systems. The **image plane coordinate system** has its origin at the image center, while the **digital image coordinate system** typically has its origin at the **top-left corner** of the image. This difference in origin is represented by the offset o_x *and* o_y .
- The image plane uses **physical units** (e.g., millimeters), whereas a digital image is represented as a grid of **discrete pixels**. The conversion from physical distance to pixel coordinates is represented by the scale factors s_x *and* s_y. s_s is included to model a system of skew coordinates (i.e. *x* and *y* axes are not perfectly perpendicular).
- Finally, **homogeneous coordinates** are used to represent the transformation as a **matrix-vector multiplication**, allowing the translation and scaling transformations to be handled conveniently in a single matrix operation.

$$\begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix} = -\frac{b}{z_c} \begin{pmatrix} x_c \\ y_c \end{pmatrix}$$

Image plane to digital image

Homogeneous coordinates

$$\begin{pmatrix} x' \\ y' \\ w \end{pmatrix} = \begin{pmatrix} -\frac{b}{s_x} & s_s & o_x & 0 \\ 0 & -\frac{b}{s_y} & o_y & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x'_c \\ y'_c \\ z'_c \\ w_c \end{pmatrix}$$

Pinhole Camera Model - Transformation from the Image Plane to the Digital Image - Extrinsic Parameters

- Since the **camera's position and orientation** may change relative to the objects, we define a fixed world coordinate frame as a common reference for describing the positions of objects independently of the camera.
- The **rotation R** and **translation t** describe the relationship between the **world coordinate frame** and the **camera coordinate frame**:

$$\begin{pmatrix} x_\mathrm{c} \\ y_\mathrm{c} \\ z_\mathrm{c} \end{pmatrix} = \mathbf{R} \begin{pmatrix} x_\mathrm{w} \\ y_\mathrm{w} \\ z_\mathrm{w} \end{pmatrix} + \mathbf{t}$$

- This completely describes the image formation process of a pinhole camera.

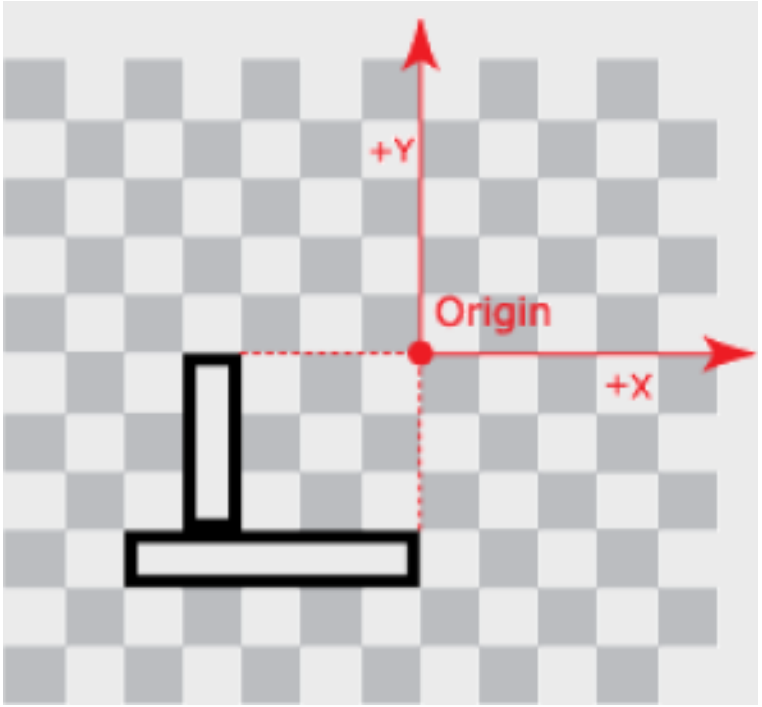
$$\begin{pmatrix} x' \\ y' \\ w \end{pmatrix} = \begin{pmatrix} -\frac{b}{s_x} & s_\mathrm{s} & o_x & 0 \\ 0 & -\frac{b}{s_y} & o_y & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} & \mathbf{R} & & \mathbf{t} \\ & & & \\ & & & \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x'_\mathrm{w} \\ y'_\mathrm{w} \\ z'_\mathrm{w} \\ w_\mathrm{w} \end{pmatrix}$$

Pinhole Camera Model

- To precisely know the transformation from the 3D real-world into 2D digital images, it requires prior knowledge of many of the camera's extrinsic and intrinsic parameters.
- If given an arbitrary camera, we may not have access to these parameters. We do, however, have access to the images the camera takes.
- How do we determine the parameters?

Pinhole Camera Model

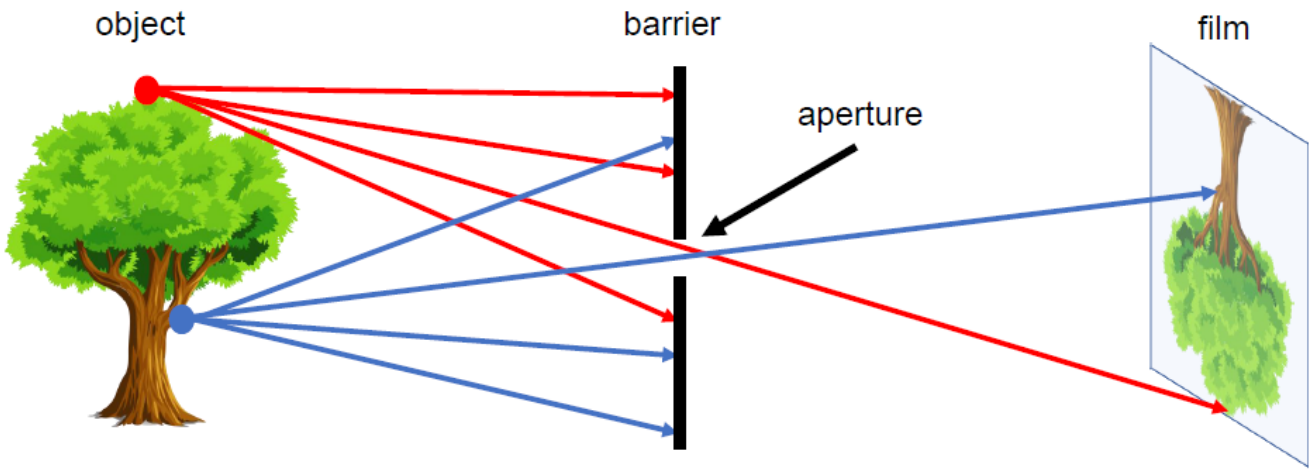
- From the **checkerboard's** known pattern, we have known points in the world reference frame.
- Finding these points in the image we take from the camera gives corresponding points in the image.
- The parameters can be determined by solving a linear system of equations from the below equation.
- Once those parameters are known, you can use the calibrated model in the **reverse direction** for measurement in machine vision:

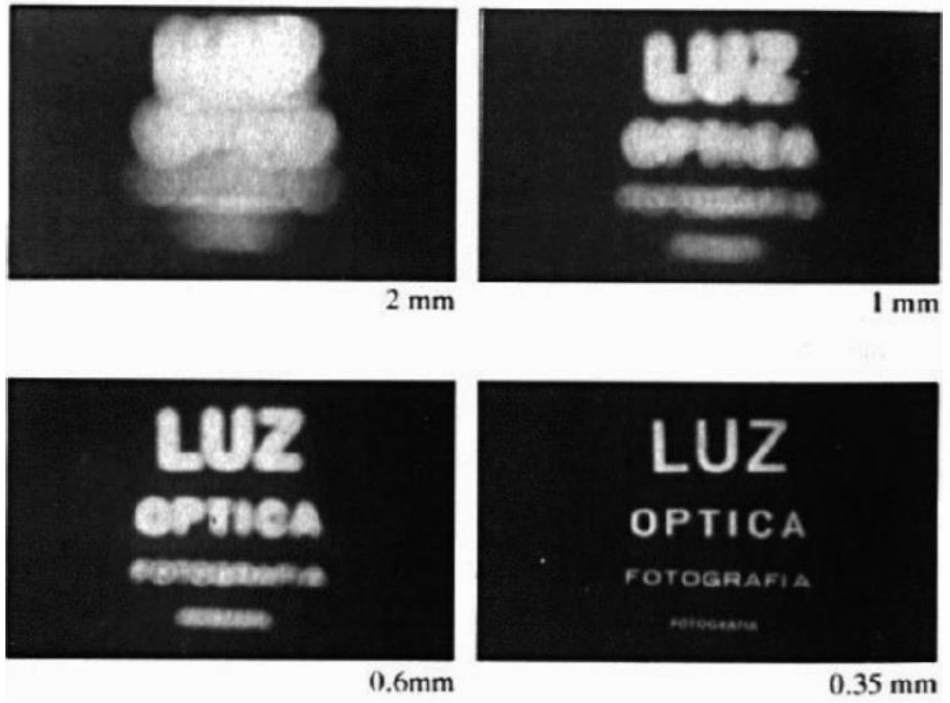


$$\begin{pmatrix} x' \\ y' \\ w \end{pmatrix} = \begin{pmatrix} -\frac{b}{s_x} & s_{\mathrm{s}} & o_x & 0 \\ 0 & -\frac{b}{s_y} & o_y & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} & \mathbf{R} & & \mathbf{t} \\ & & & \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x'_{\mathrm{w}} \\ y'_{\mathrm{w}} \\ z'_{\mathrm{w}} \\ w_{\mathrm{w}} \end{pmatrix}$$

Pinhole Camera Model

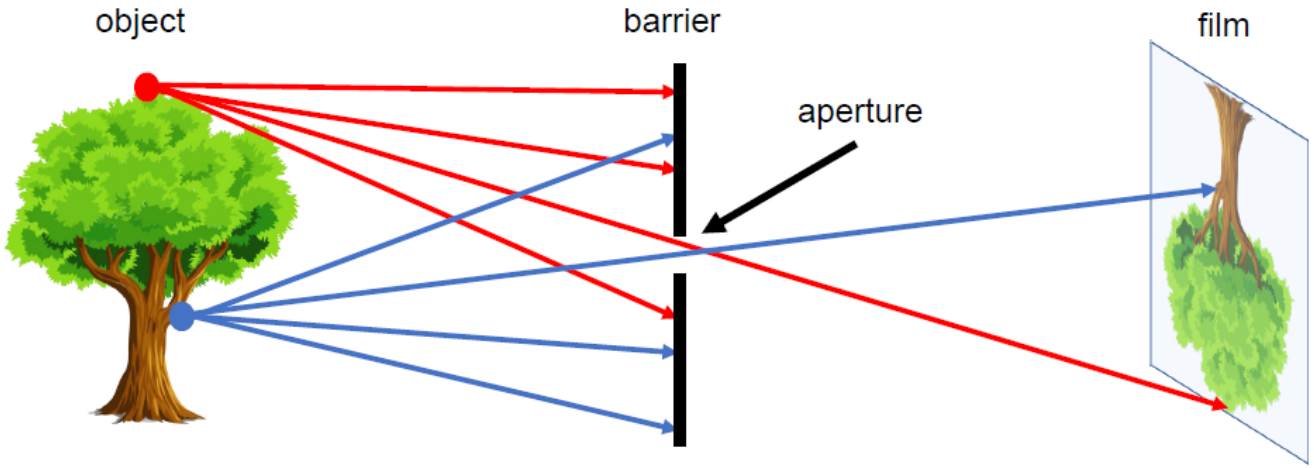
- Effects of the aperture size on the image quality:
- As the aperture size increases, the number of light rays that passes through the aperture consequently increases, then each point on the film may be affected by light rays from multiple points in 3D space, blurring the image.

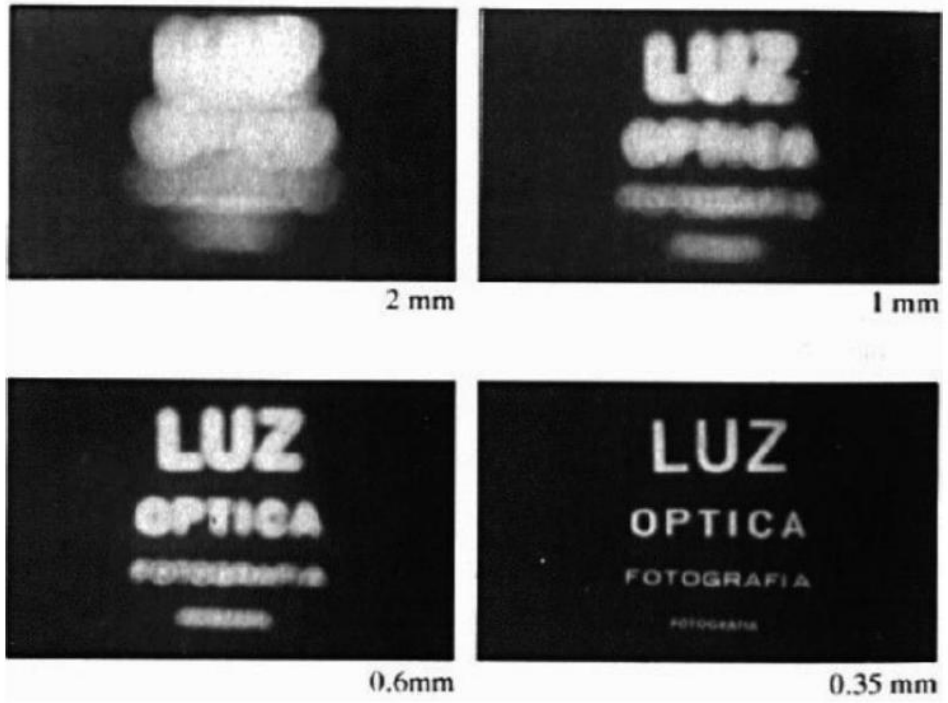




Pinhole Camera Model

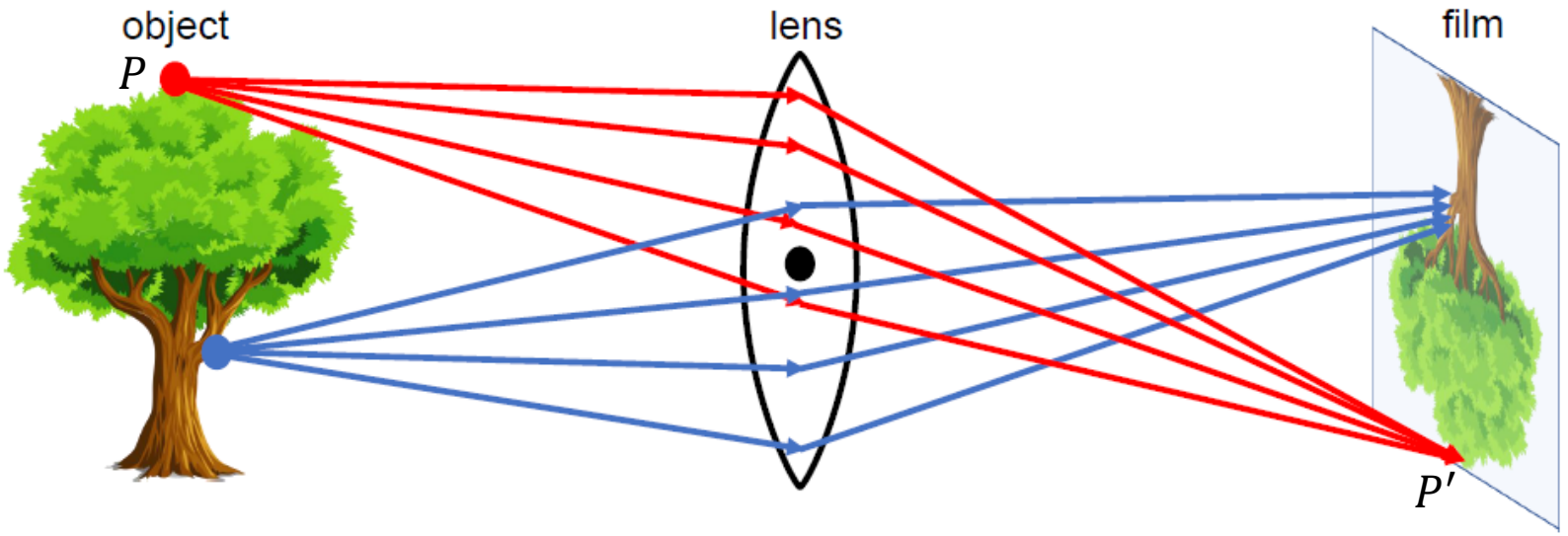
- But a smaller aperture size causes fewer light rays to pass through, resulting in crisper but darker images.
- Therefore, we arrive at the fundamental problem presented by the pinhole formulation: can we develop cameras that take crisp and bright images?





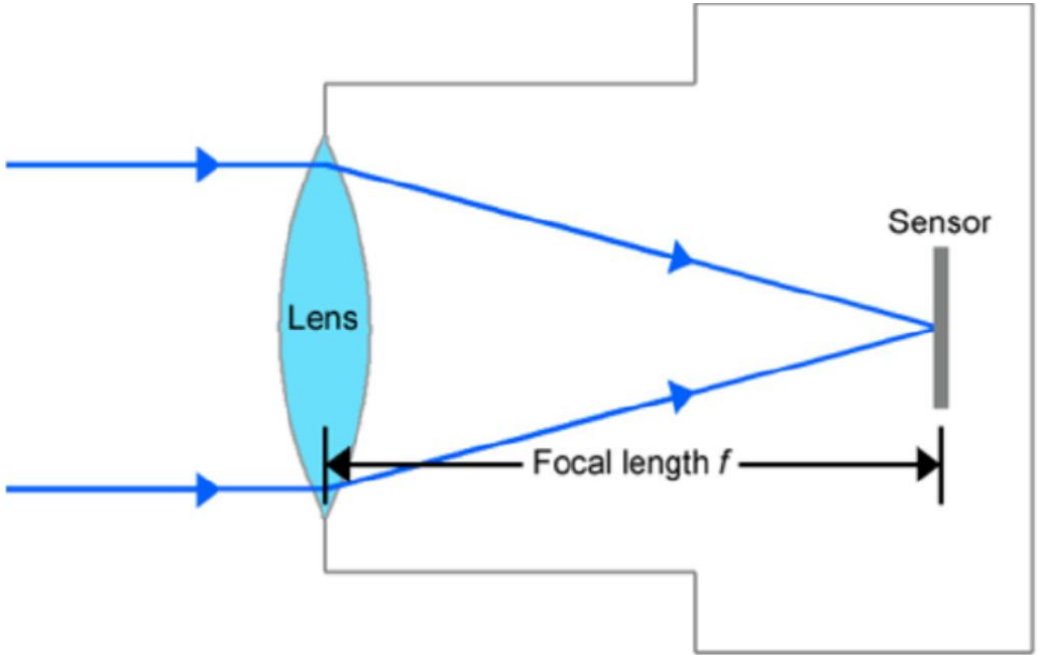
Camera Model with Lens

- **Camera Model with Lens**: Use lenses to gather more light and focus it onto the sensor, allowing for brighter and sharper images.



Camera Model with Lens

- **Focal Length** (mm) of a lens is the distance from the center-point of the lens to the point at which light rays parallel to the center-line of the lens are focused.
- The shorter the focal length, the more the lens bends the light rays, the wider its angle of view.
- For example, a lens with a focal length of 28mm can fit a lot more of an object into the field of view than can a lens with a focal length of 200mm.
- The focal length of a lens is also considered an expression of its magnification power.
- For example: A lens with a range of focal length from 5.8mm to 17.4mm is called a 3x zoom, because 17.4 = 3 x 5.8.

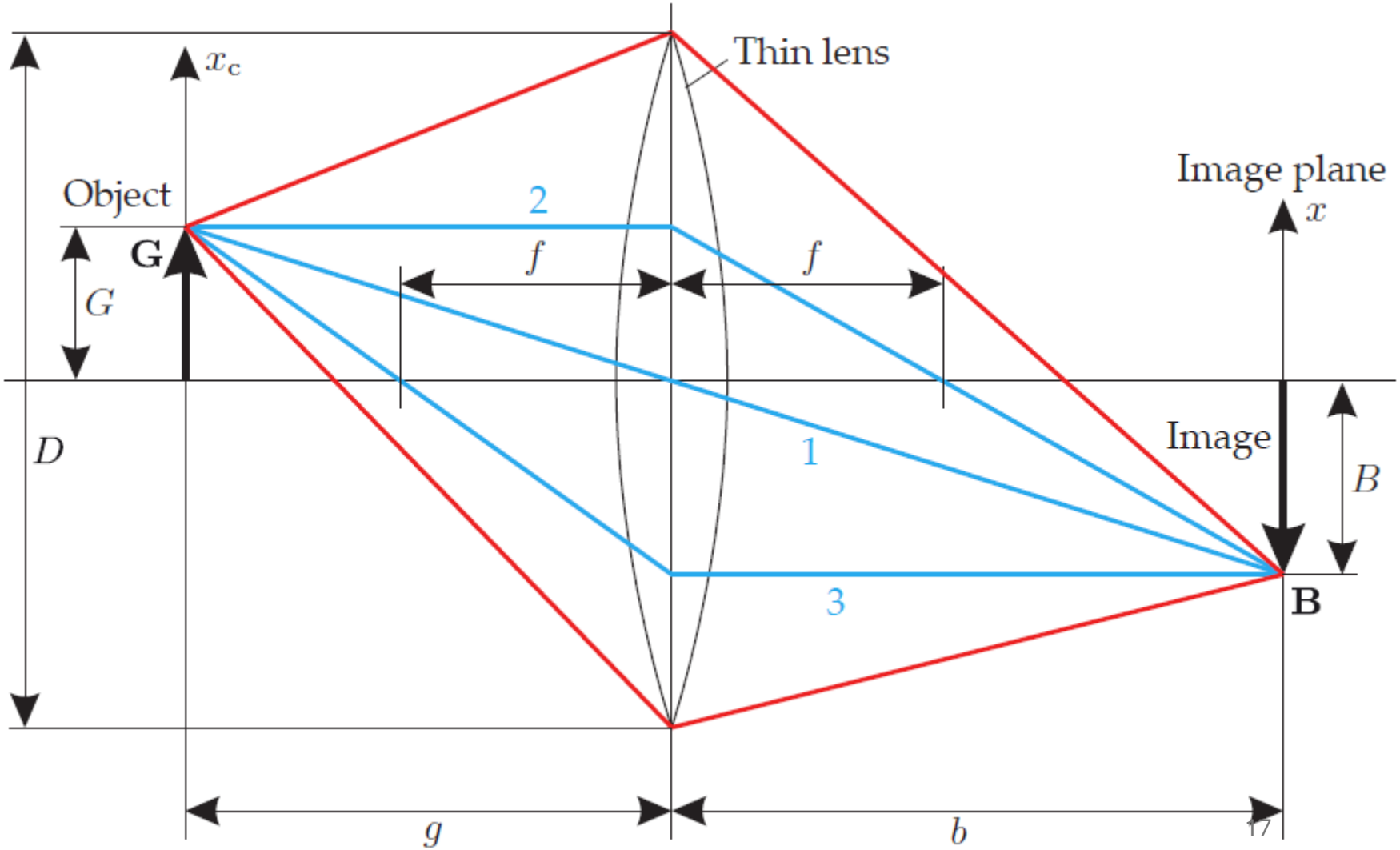


Camera Model with Lens

$$\frac{G}{f} = -\frac{B}{b-f} \qquad \frac{G}{g} = -\frac{B}{b}$$

- For a focused optical imaging with object distance of *g*, an image distance of *b* and a lens with focal length *f*, the **lens equation** is:

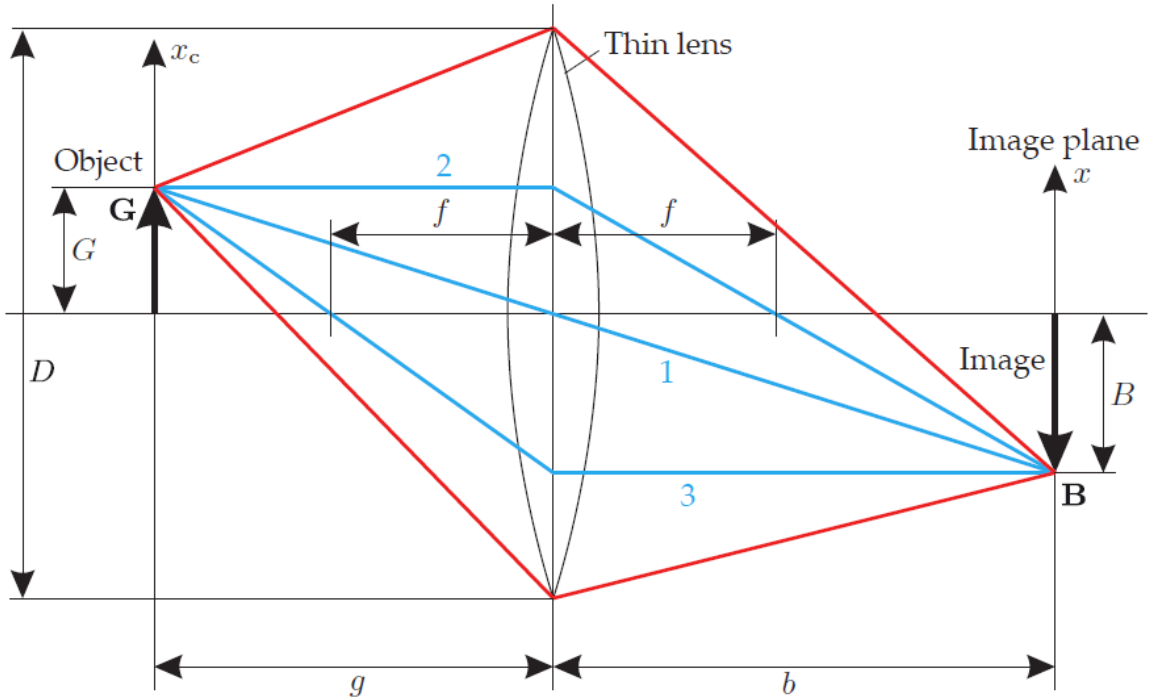
$$\frac{1}{f} = \frac{1}{g} + \frac{1}{b}$$



Camera Model with Lens

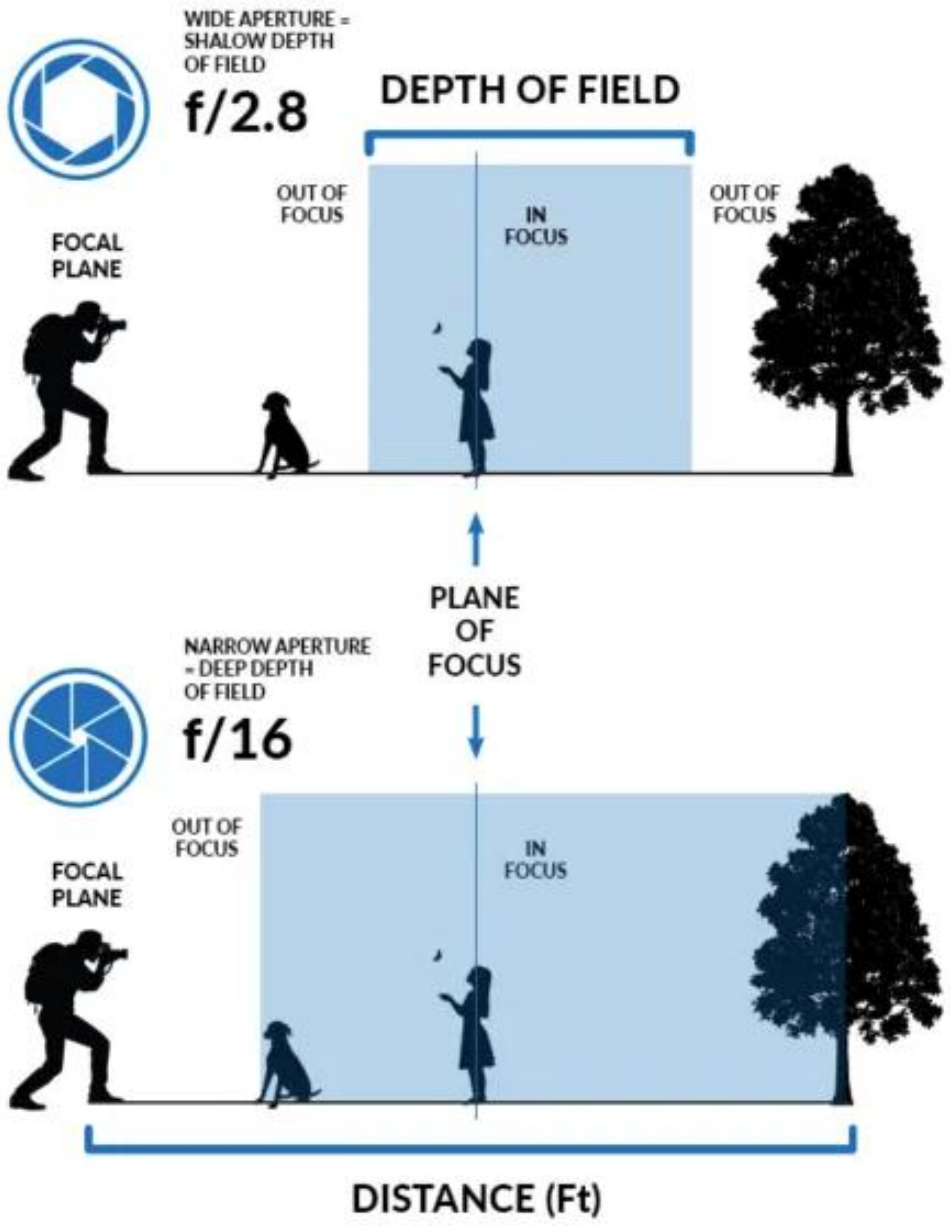
- According to the lens equation, exactly one object plane perpendicular to the optical axis is captured sharply on the image plane.
- All objects that are not located exactly at that distance from the camera will appear blurred.
- As the dimensions of the light sensitive elements (pixels) on an image sensor are finite (not infinitely small), a certain blur corresponding to a specific range of object distances can be tolerated.
- For example, a perfectly focused point produces a spot size of $1\ \mu m$. The camera pixel size is $5\ \mu m$.
- The pixel already averages light over a $5\ \mu m$ area. Whether the light spot is $1\ \mu m$ or $3\ \mu m$, the sensor records nearly the same value.
- Therefore, slight defocus becomes invisible.

$$\frac{1}{f} = \frac{1}{g} + \frac{1}{b}$$



Depth of Field

- The depth of field of an optical system is the length of the interval with respect to z_c containing objects which are captured with tolerable sharpness.



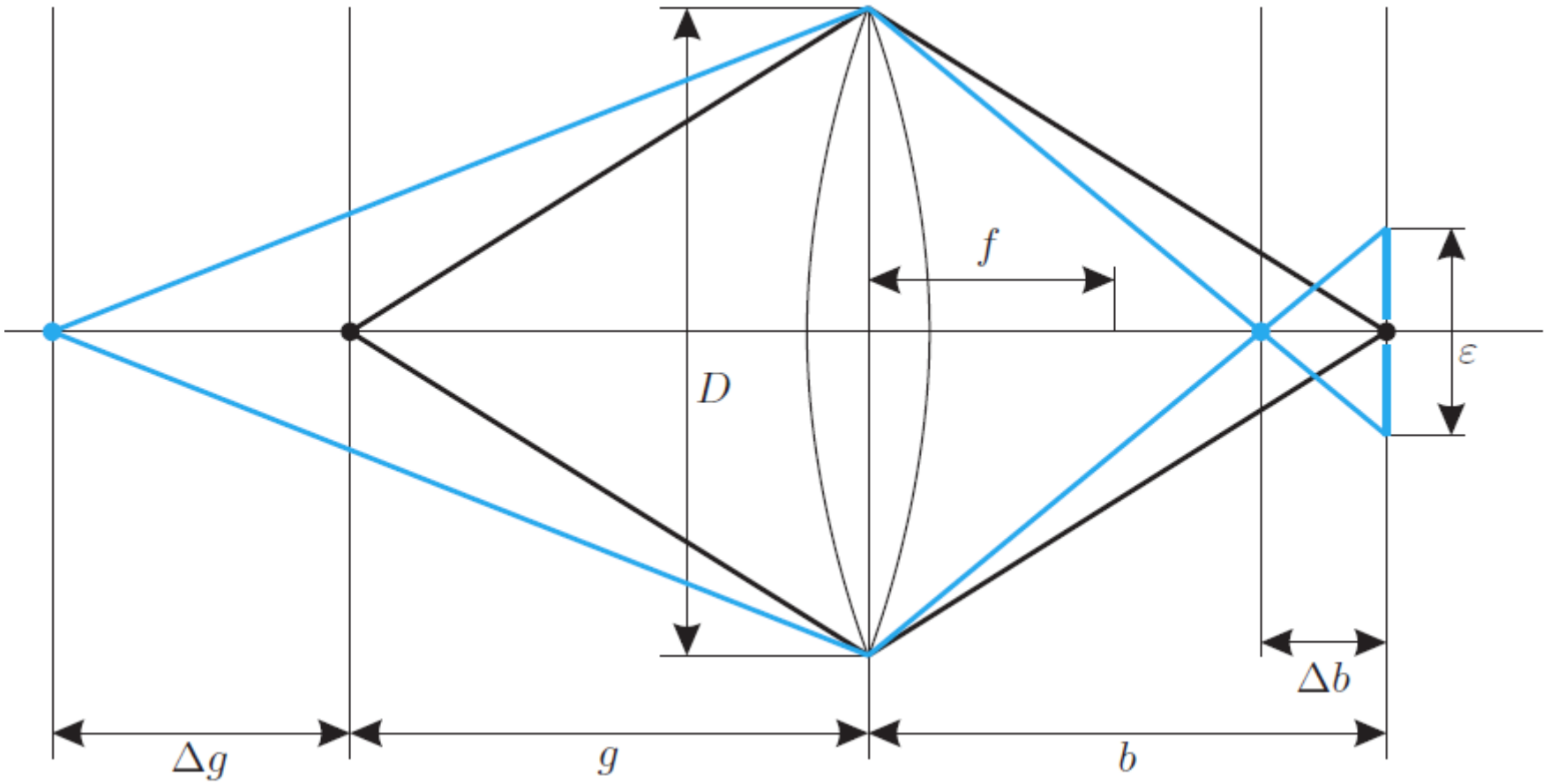
Depth of Field

- The formula for Δg :

$$\Delta g = \Delta g(\varepsilon) = \frac{\varepsilon\, g}{\frac{D\, f}{g-f} - \varepsilon}$$

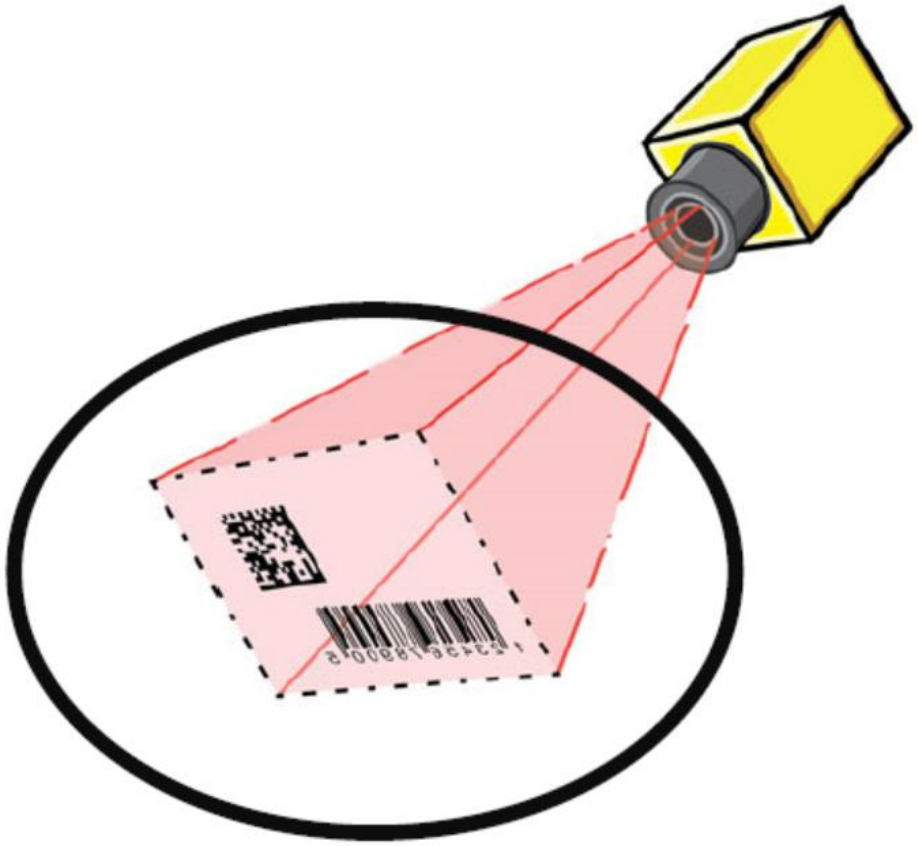
- The depth of field surrounding the object plane g:

$$\Delta g_{\text{far,near}} \approx \pm \varepsilon_{\max} \frac{g\,(g-f)}{D\, f}$$



Field of View

- **Field of View** is the imaged area of an object under inspection. This is the portion of the object that fills the camera's sensor.
- Field of View is critical for choosing the correct optical components to use in an imaging application. Since resolution is dependent of field of view, determining field of view affects what one is trying to analyze or measure.



Resolution vs. Field of View

- **Resolution**: Number of pixels in the sensor (e.g., 640 x 480). Higher resolution captures more detail.
- What can a camera with more pixels do for your application?
- It can provide more pixels per feature for better accuracy (same FOV).

Same Field of View, more pixels per feature

800 x 600



1600 x 1200

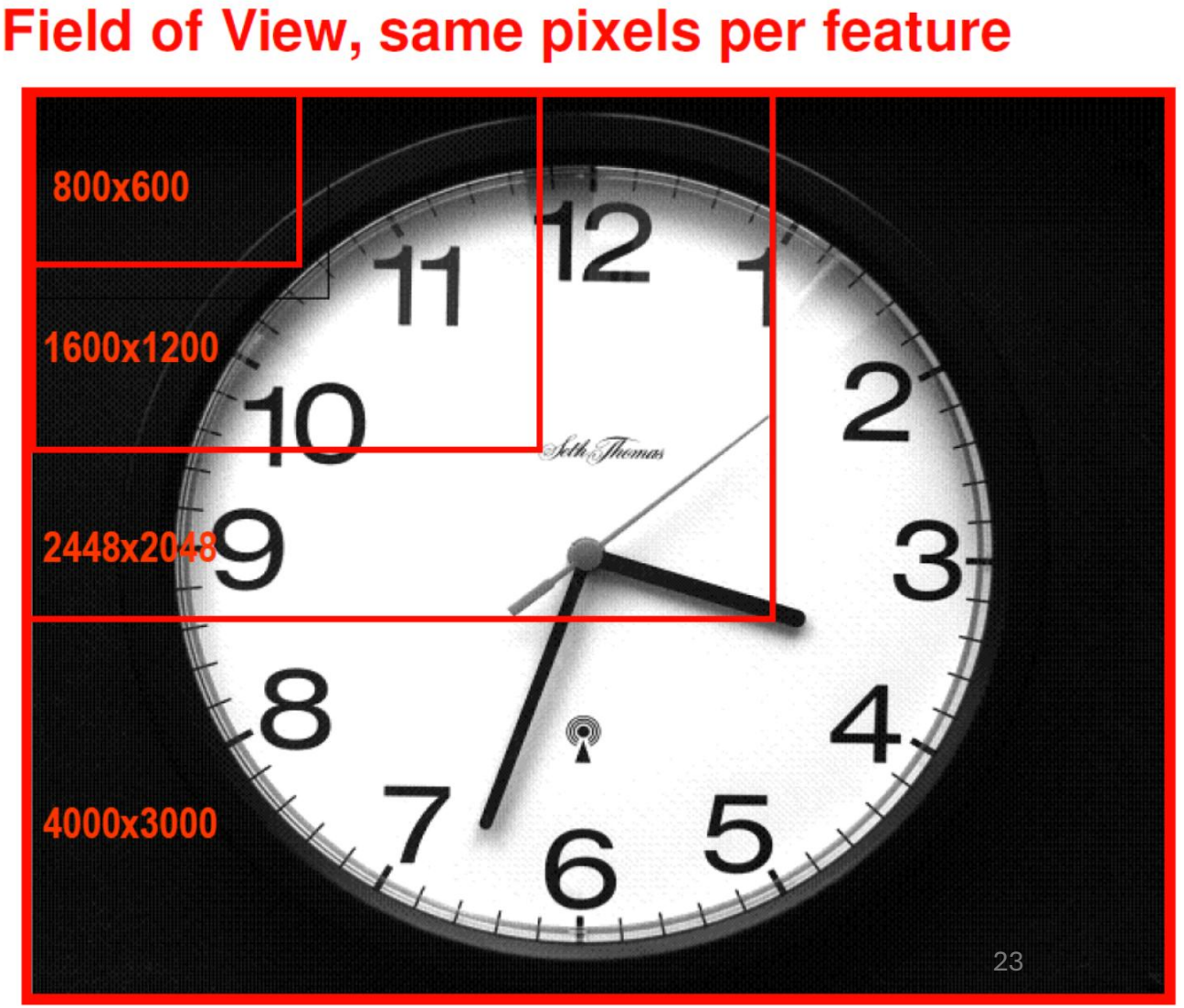


4000 x 3000



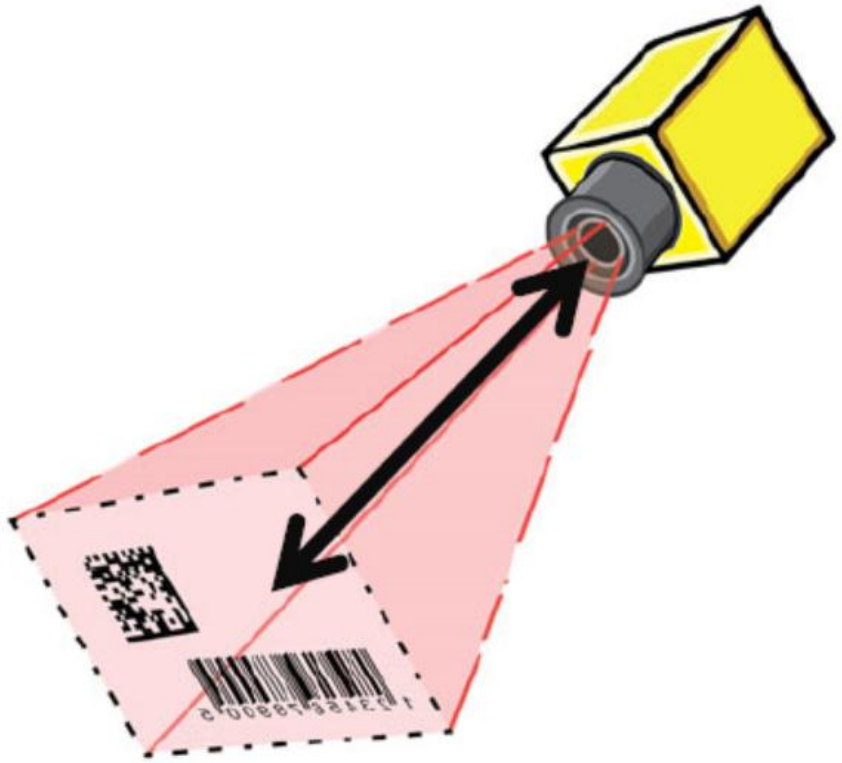
Resolution vs. Field of View

- This example of a clock shows how **four cameras of different resolution** can obtain the same pixels per feature at increasing Fields of View.
- A 640x480 resolution camera can only show a portion of the clock. A 1600x1200 camera can show more of the clock, because it can be moved further away (larger Field of View).
- The higher the resolution of the camera, the more of the clock that can be viewed at the same pixels per feature.



Working Distance

- **Working Distance** is the distance from the front of the lens to the object. Working distance determines how far to position an imaging lens from the object under inspection.
- Working distance is critical when considering space constraints and lighting geometry.



Field of View, Focal Length, & Working Distance

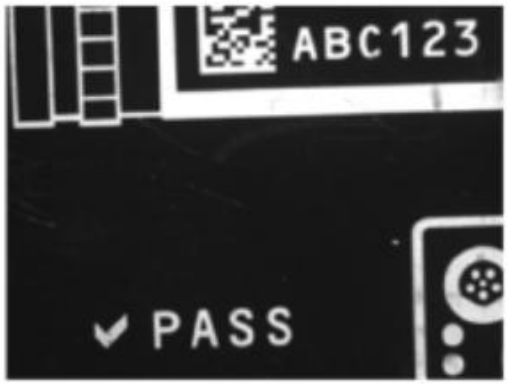
- If you're focused on a part and need **larger** FOV: Go to a smaller focal length lens, for example 25mm -> 16mm.
- If you're focused on a part and need to mount camera **closer** (WD): Go to a smaller focal length lens, for example 25mm -> 16mm.
- The smaller the focal length, the closer the WD and larger the FOV.



12mm



16mm



25mm

Contrast

- **Contrast** describes how well **differences in brightness**, such as black and white, can be distinguished at a given resolution.
- The **lens**, **sensor**, and **lighting** all play important roles in determining the contrast of the resulting image. If any of these components is not properly selected or configured, it can reduce the overall contrast of the imaging system at a given resolution.

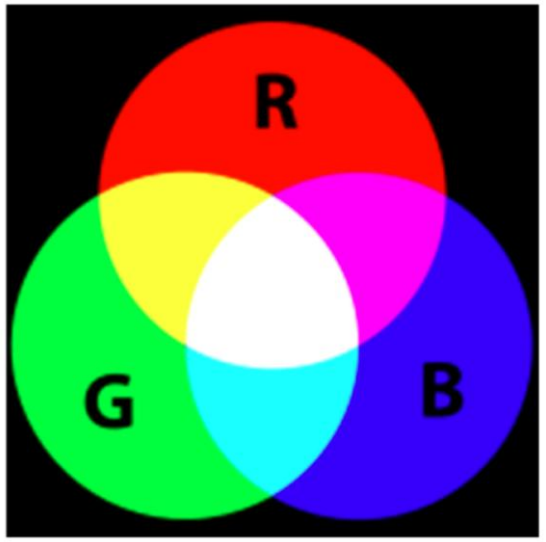


Pixel

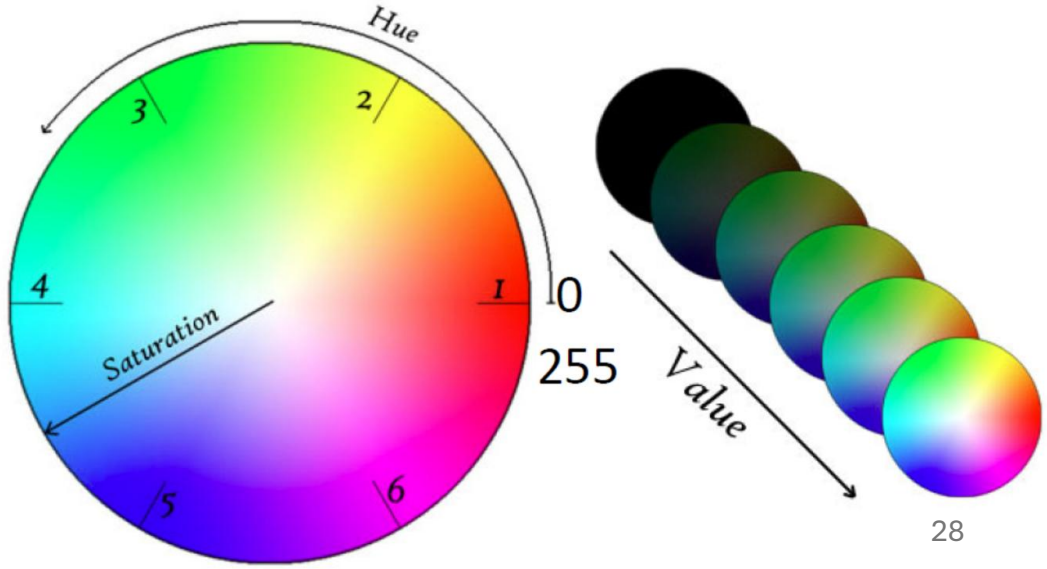
- A **pixel** (short for **picture element**) is the **smallest addressable element of a digital image**. It represents a single location in the image and stores **brightness or color information**. When thousands or millions of pixels are arranged in a grid, they form a digital image that can be displayed, analyzed, or processed.
- Each pixel has a **location within the image**, represented by its **X/Y coordinates**, and stores a **brightness or color value**.
- In a **monochrome camera**, the brightness of each pixel is represented by a **grayscale value** from **0 to 255**, where **0 represents black** and **255 represents white**.



Color Space

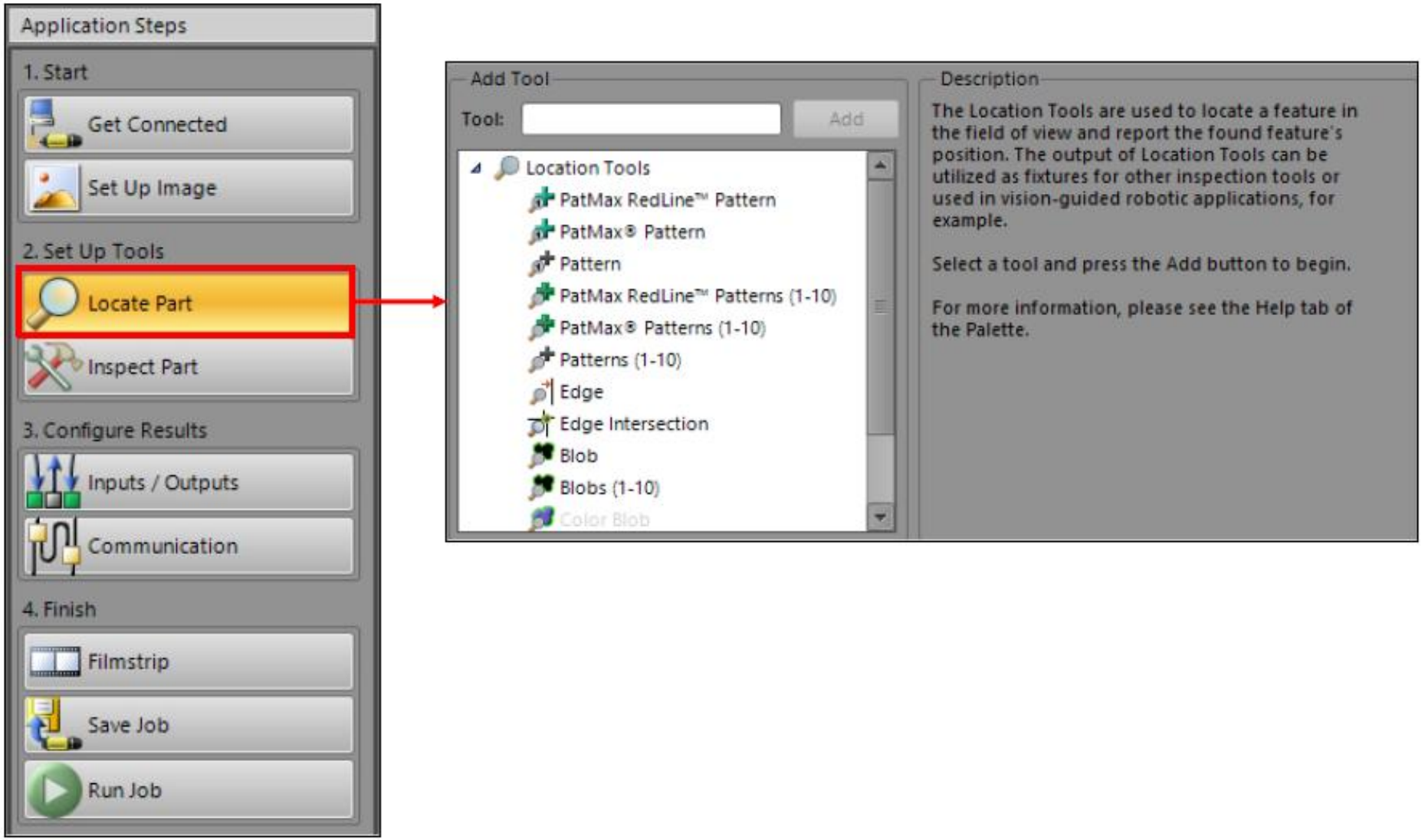


- In a **color image**, each pixel is represented by **three color components**. In an 8-bit representation, each component is typically assigned a value from **0 to 255**.
- These three components can be represented using different **color spaces**, such as:
  - **Red, Green, and Blue (RGB)**
    - Red = **(255, 0, 0)**
    - Green = **(0, 255, 0)**
    - Blue = **(0, 0, 255)**
  - **Hue, Saturation, and Value (HSV)**
    - Red = **(0, 255, 85)**
    - Green = **(85, 255, 85)**
    - Blue = **(170, 255, 85)**



Location Tools

What Are Location Tools

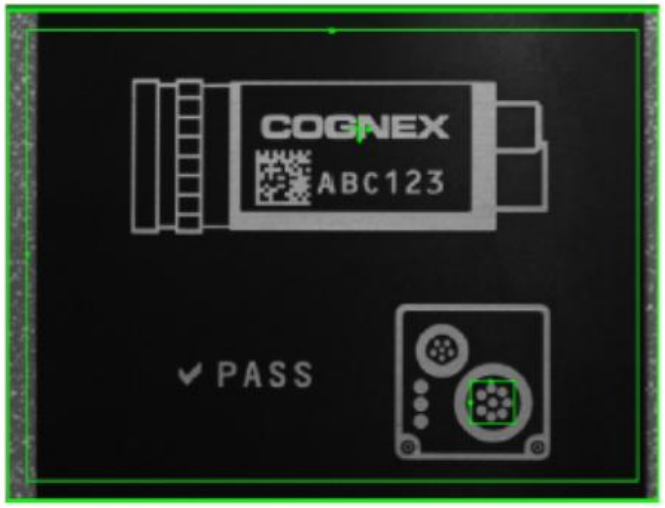


What Are Location Tools

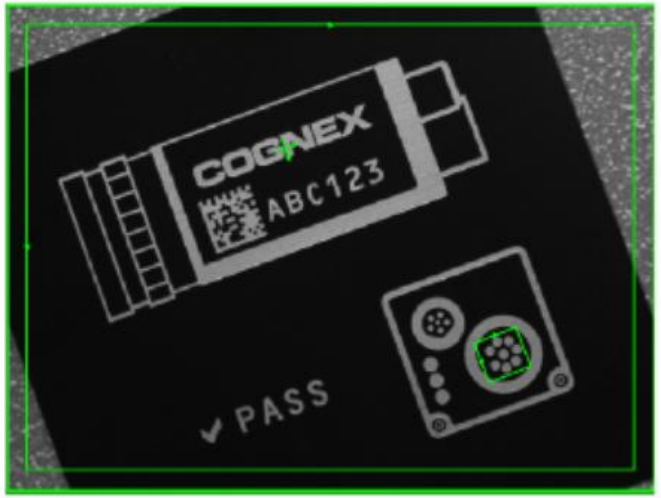
- **Location Tools** are used to define a **feature** in the image that provides positional data. They create a **Fixture**, which is used to locate a part in the image quickly and reliably, even if the part being inspected rotates or appears in different locations in the image.

How Fixture Works

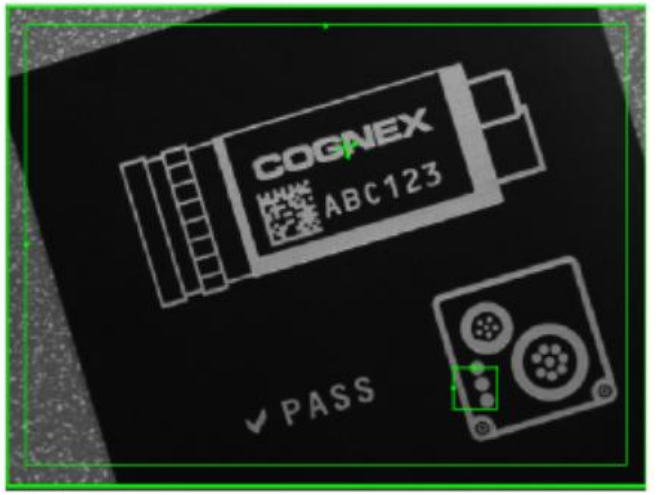
- When a **feature** is used as a **fixture** during an inspection, the In-Sight camera will first search for the feature.
- Once the feature has been located, the feature's position serves as a reference point for the other vision inspection tools, which use the position of the feature to orient their regions.



Initial Set Up



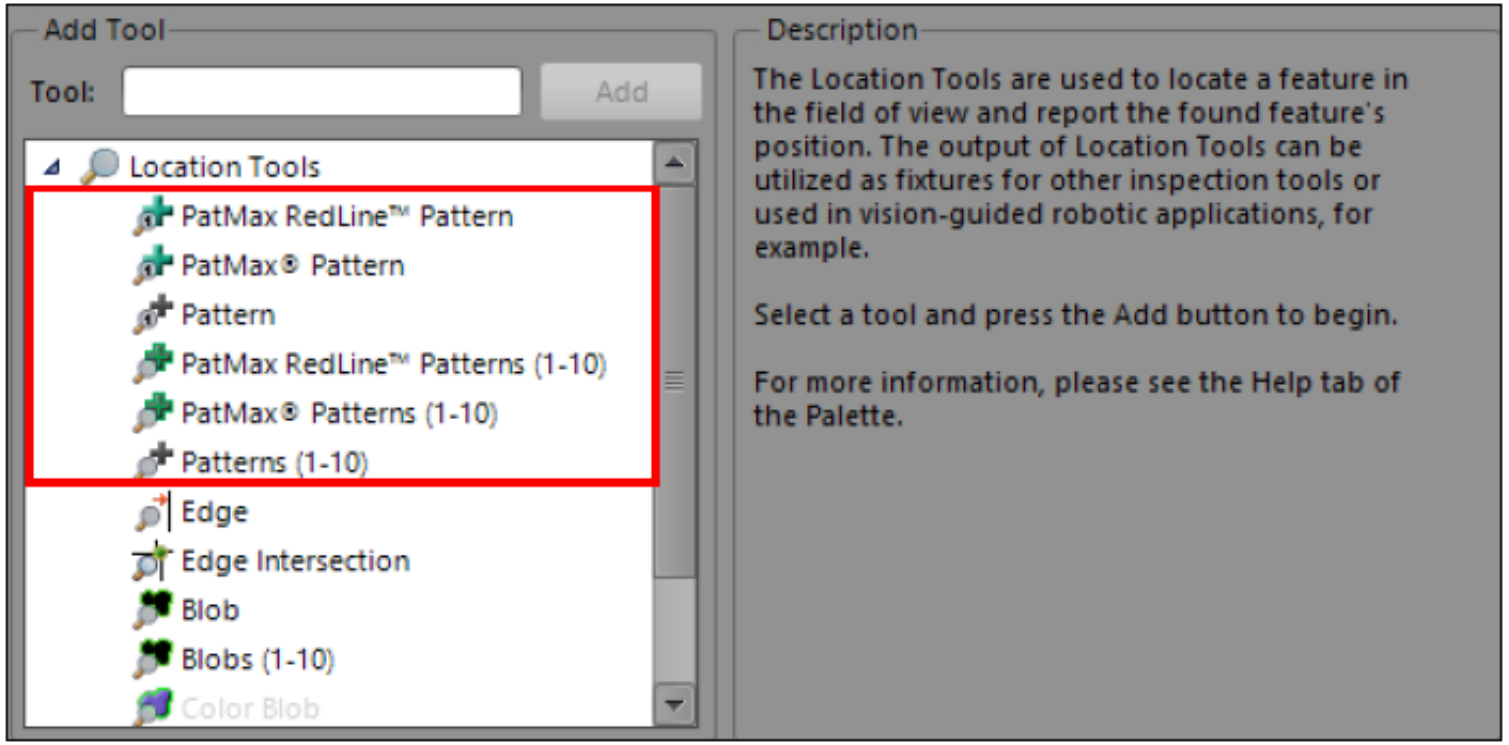
Fixtured



Not Fixtured

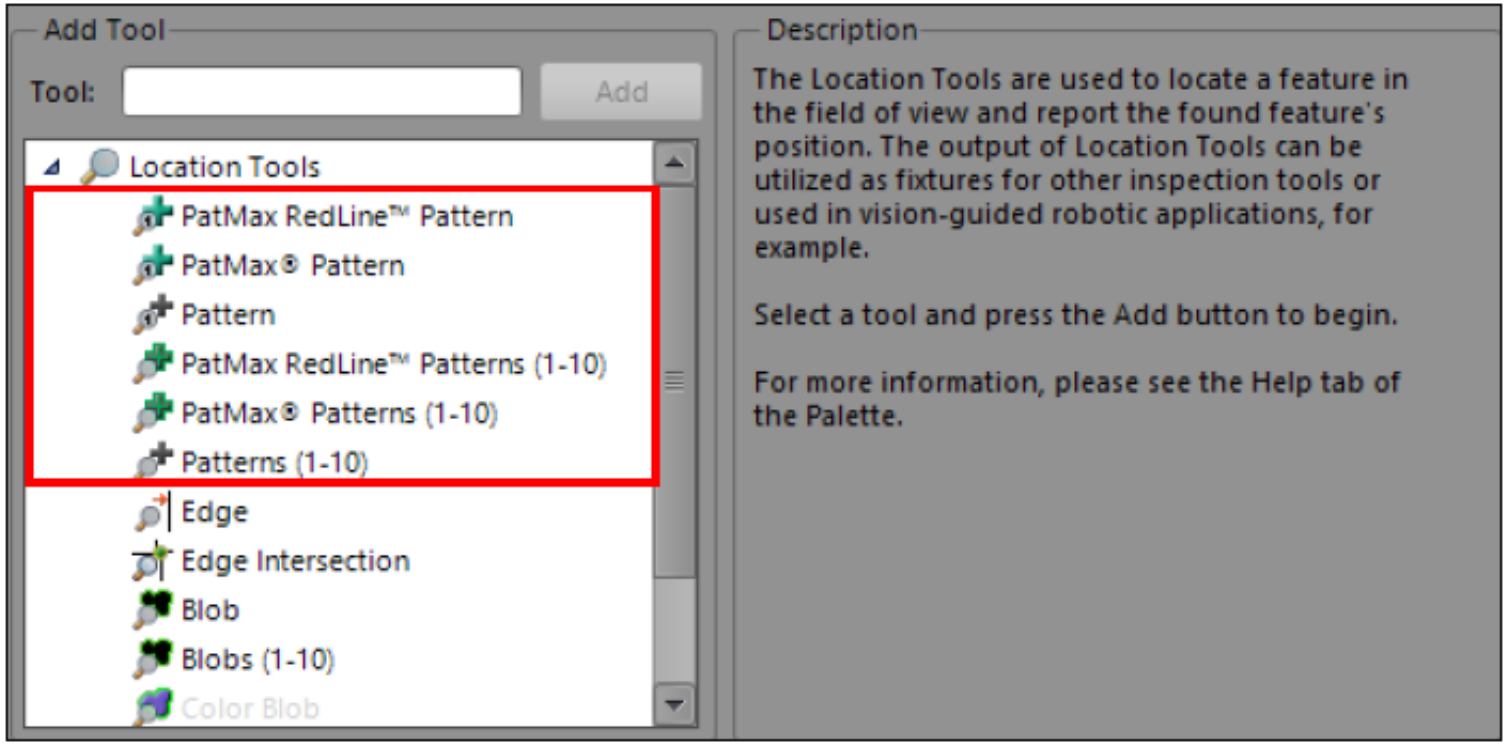
Pattern Matching Tools

- The **Pattern Matching** tools locate one pattern feature, report the X, Y coordinates, angle and score of the found pattern.
- It is commonly used as a fixture to orient other vision tools.
- If more than two pattern features need to be found, **1–10 Pattern tools** should be used.



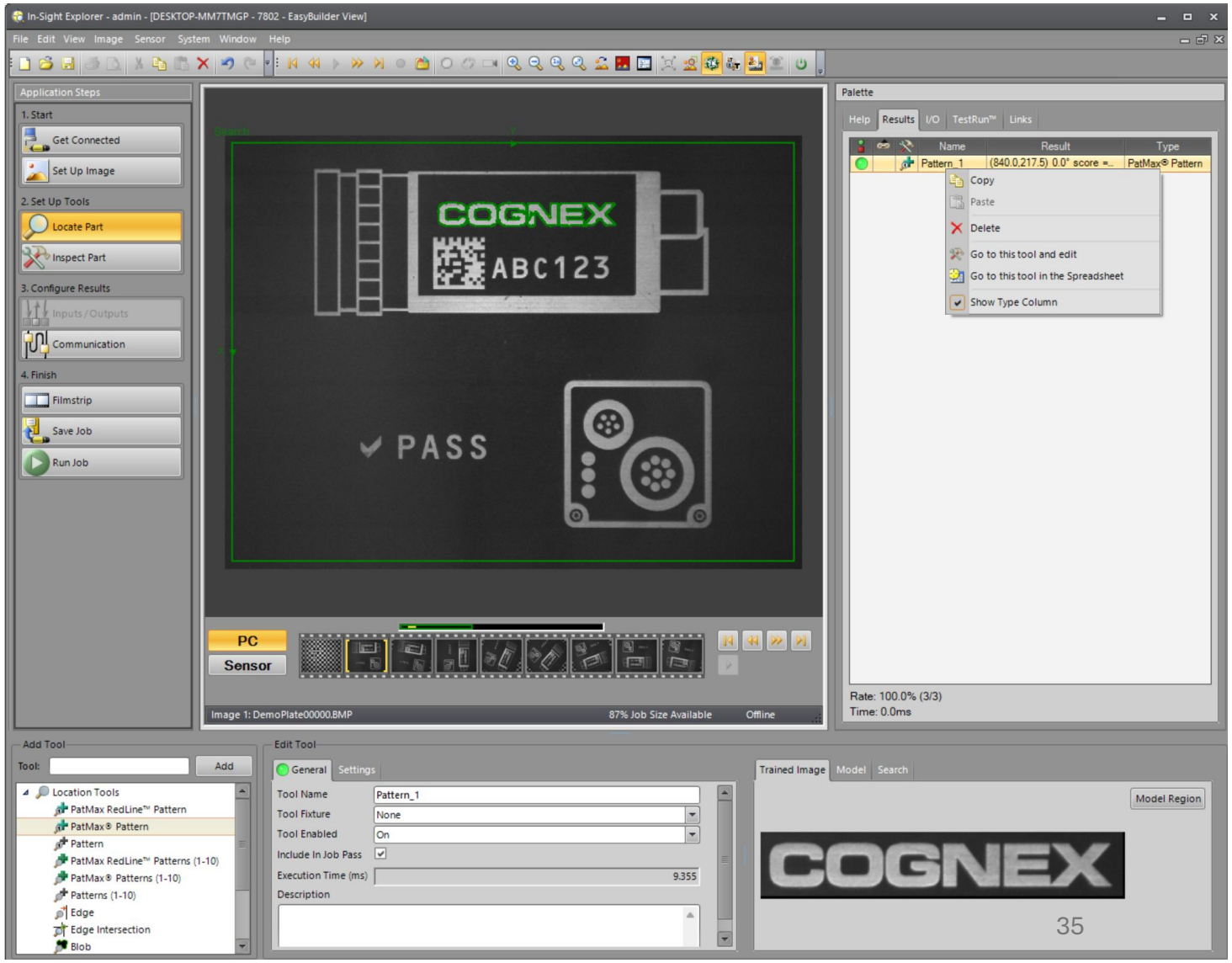
Pattern Matching Tools

- To select a tool to your job, select the appropriate tool from the **Add Tool** group and press the **Add** button, or double click on the tool. You will then be prompted to define the feature in the image.



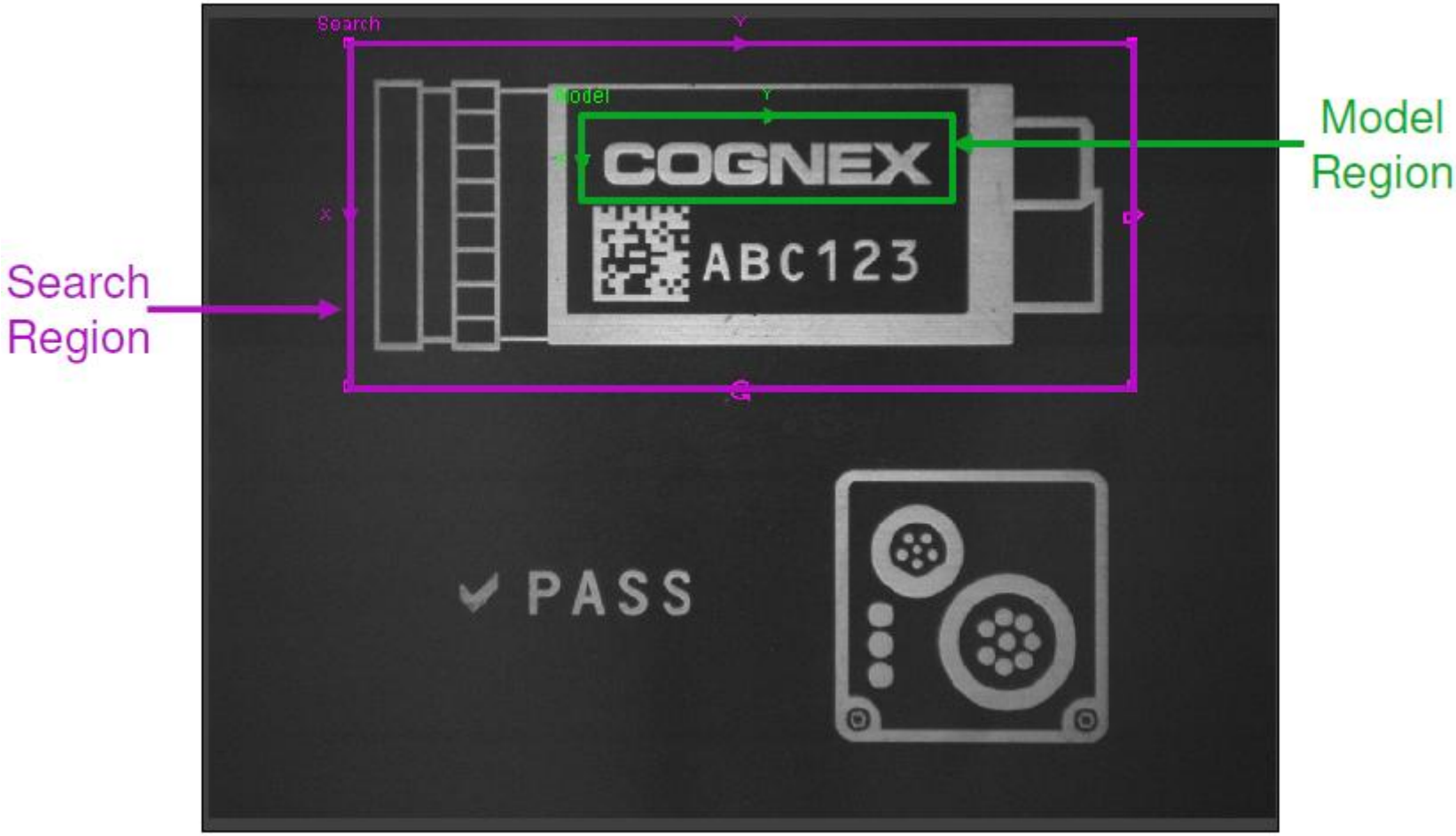
Pattern Matching Tools

- To **Delete** a tool from your job, simply do one of the following:
- Select the tool in the **Palette ->** Results table and press the <Delete> key on the keyboard or press the Delete icon in the Standard toolbar.
- Right-click on the tool in the Results table and select the Delete option from the menu.



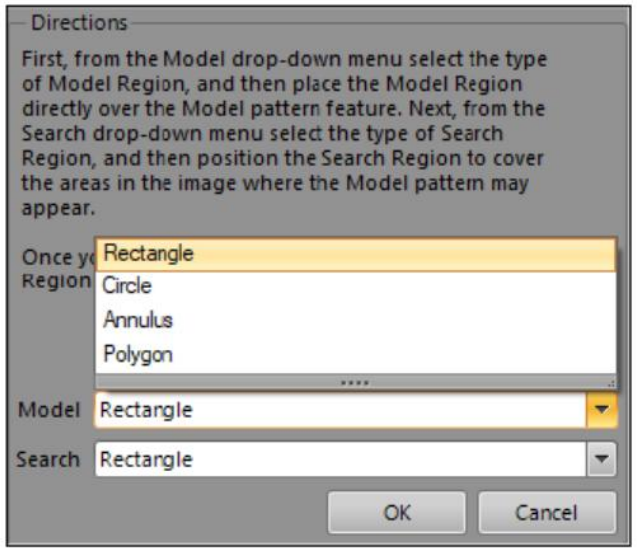
Pattern Matching Tools

- When adding a Pattern Matching Tool, two regions need to be defined – **Search Region** and **Model Region**.
- Regions define the area of the image where the Tool will perform its operation and are also referred to as the Region of Interest.

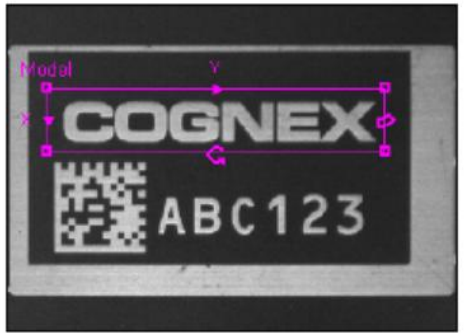


Setting Regions

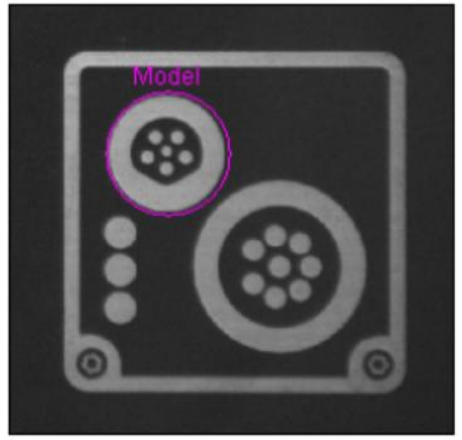
- There are four types of regions used by the Location Tools:
- **Rectangle** – the most common.
- **Circle** – are used to define circular features.
- **Annulus** – are used to define circular edge features.
- **Polygon** –are used to define irregular features by constructing a shape out of line segments.



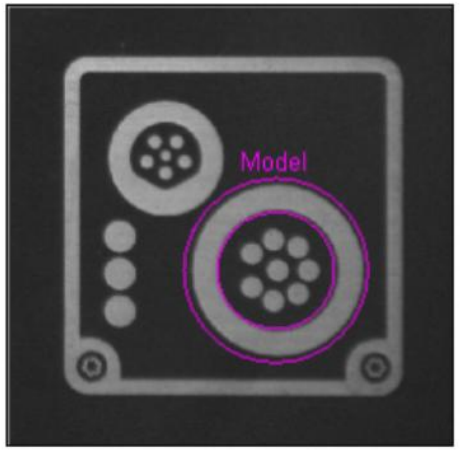
Rectangle



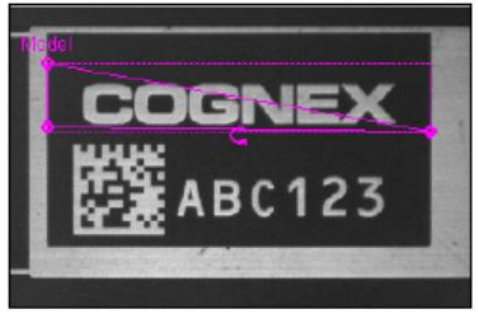
Circle



Annulus

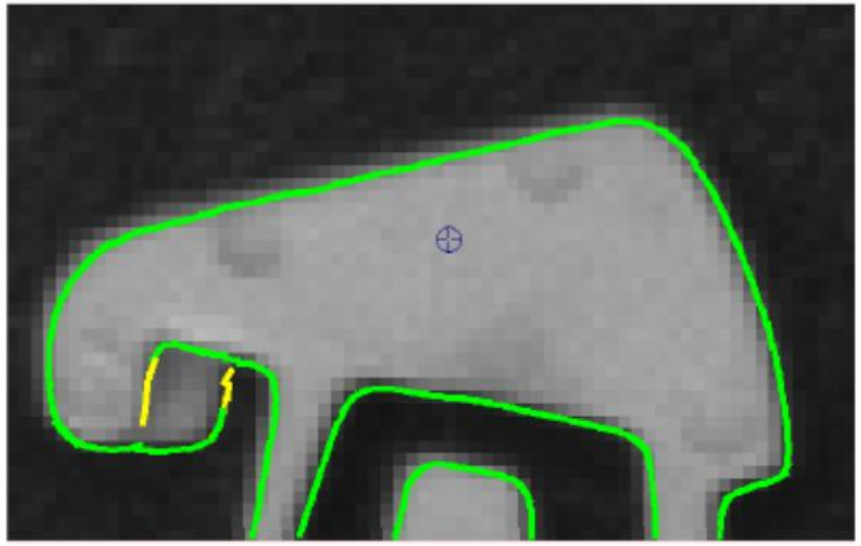


Polygon



What Is PatMax?

- A **PatMax** pattern is a collection of **geometric features**, where each feature is a point on the boundary between two regions with **different pixel values**.
- **PatMax** calculates a **geometric representation** of the object's boundaries and curves.
- **PatMax** is a feature-based pattern search technology.



PatMax: Feature-based

Find the Features - Theoretical Foundations

- Pixel intensity on an image: $I(x, y)$, x and y represent the pixel location
- Gradient:

$$\nabla I = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}\right)$$

- Gradient Magnitude

$$| \nabla I | = \sqrt{I_x^2 + I_y^2}$$

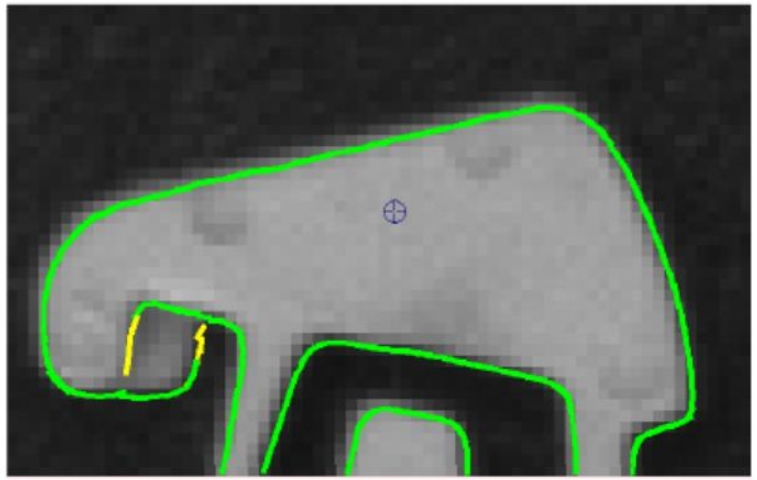
- Orientation:

$$\theta = \tan^{-1}\left(\frac{I_y}{I_x}\right)$$

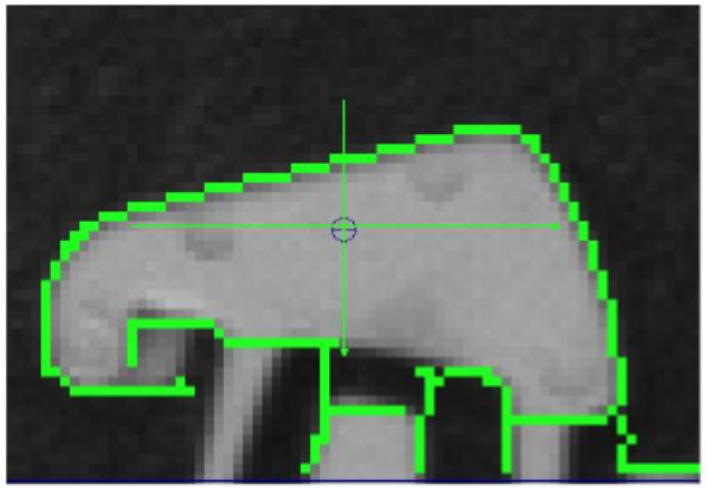
- The **gradient** tells you, how fast and in what direction intensity is changing.
- For example, edges on the image correspond to **large gradient magnitudes.**

PatMax vs. FindPatterns

- **PatMax** is more accurate than **FindPatterns** because PatMax uses geometric representations of features, while FindPatterns approximates curves using **line segments that follow the pixel grid**. FindPatterns does not efficiently and accurately handle **rotation and scaling**.
- **FindPatterns** is a pixel grid-based pattern search technology.



PatMax: Feature-based



FindPatterns: Pixel grid-based

Key Features of PatMax

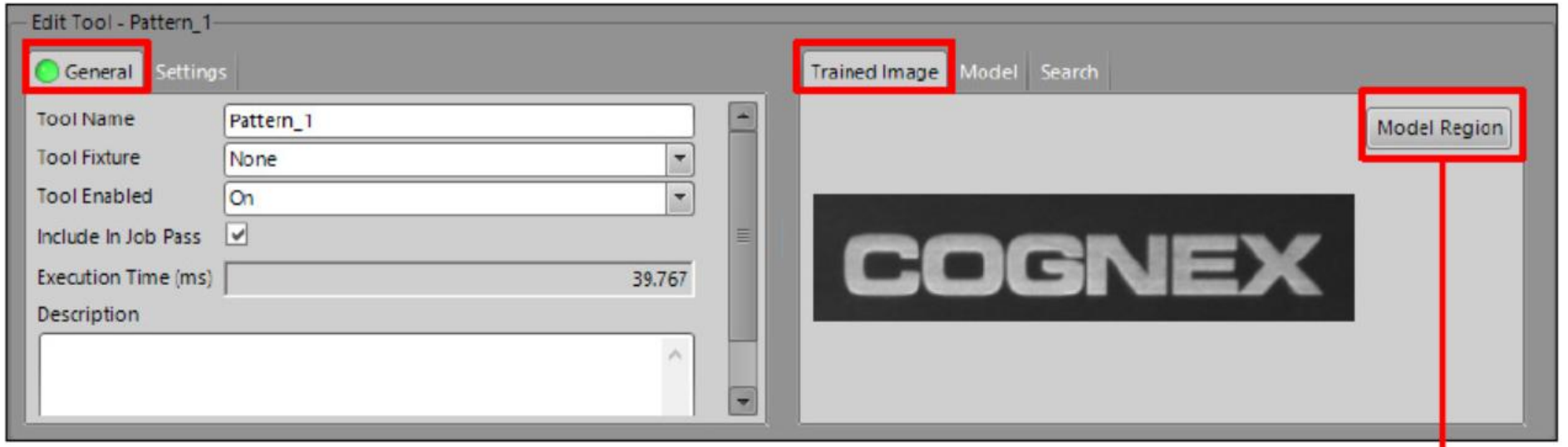
- PatMax offers three key features that distinguish it from other pattern-location technologies in machine vision:
  - **High-speed location** of objects whose appearance is rotated, scaled, and/or stretched
  - **Shape-based location** that relies on the object's shape rather than greyscale values
  - Very high accuracy

When to Use PatMax?

- Situations that require the accuracy and reliability of the PatMax Patterns Tool:
  - When variations in **lighting and reflections** are difficult to control. For example, metal parts can reflect light in different directions.
  - When the pattern being inspected has a **similar shape or shading to objects in the background**, or when the pattern is **overlapped or partially occluded** by other objects in the image.
  - When you need to **accurately distinguish one type of pattern from other similar patterns**.
  - When the conditions of the **deployment environment** are too demanding for the **Patterns Location Tool** to provide **consistent and reliable results**.

PatMax Pattern

- The **General** tab is generic for all location tools. Some of the parameters include tool name, fixture tool to use, execution time (ms) and a description.

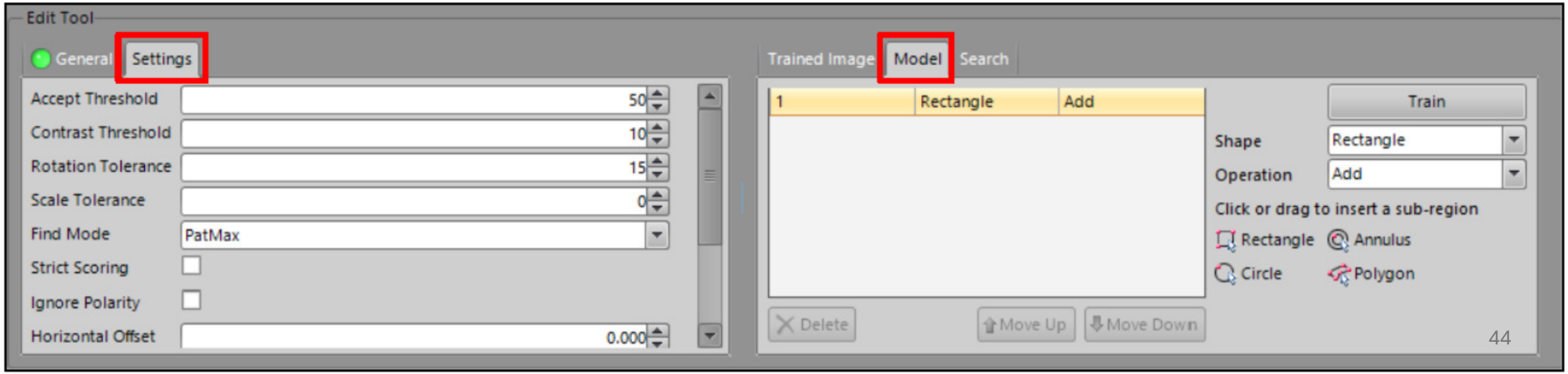


- The image displayed in the Trained Image tab is a thumbnail of the Model Pattern.
- Click the **Model Region** button to resize or move the Model region, and then click the Train button to retrain the Model pattern.

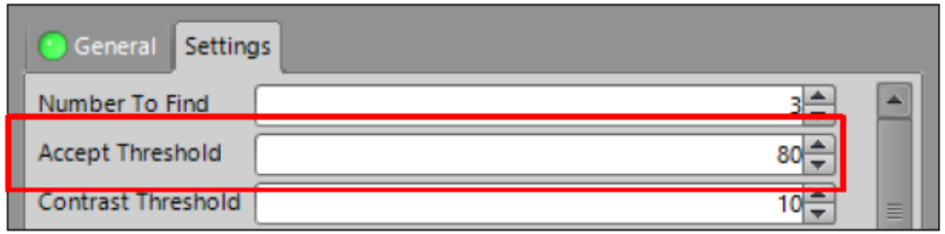


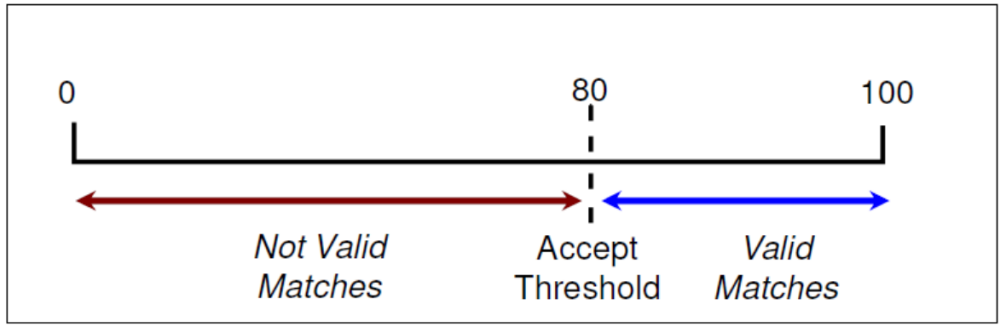
PatMax Pattern

- The **Model** and **Search** tabs are used to configure the **composite regions** for the Model and Search areas of the **PatMax Pattern** tool.
- A **composite region** is a collection of **sub-regions** (Rectangle, Annulus, Circle, or Polygon) that either **Add to** or **Subtract from** the region:
  - **Add to the region**: The sub-region is included in the image analysis.
  - **Subtract from the region**: The sub-region is excluded from the image analysis, also known as **masking**.

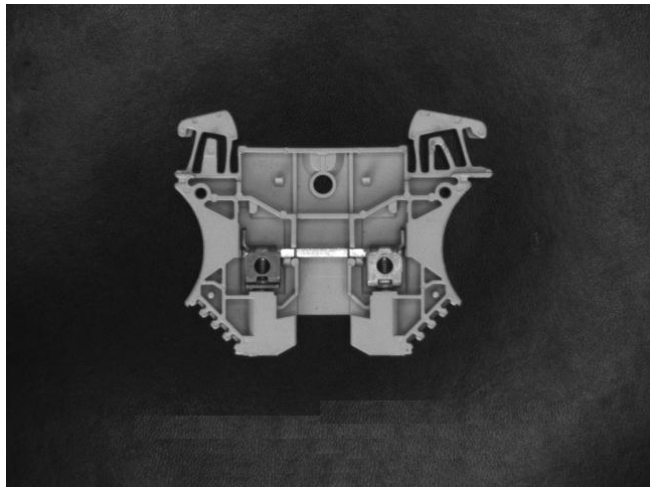


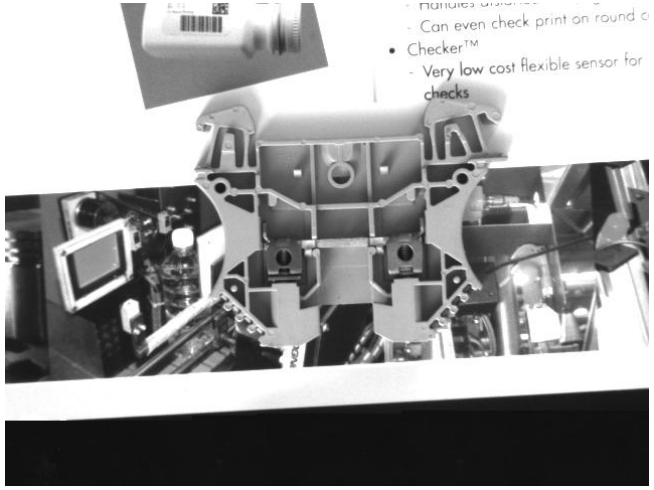
PatMax Pattern





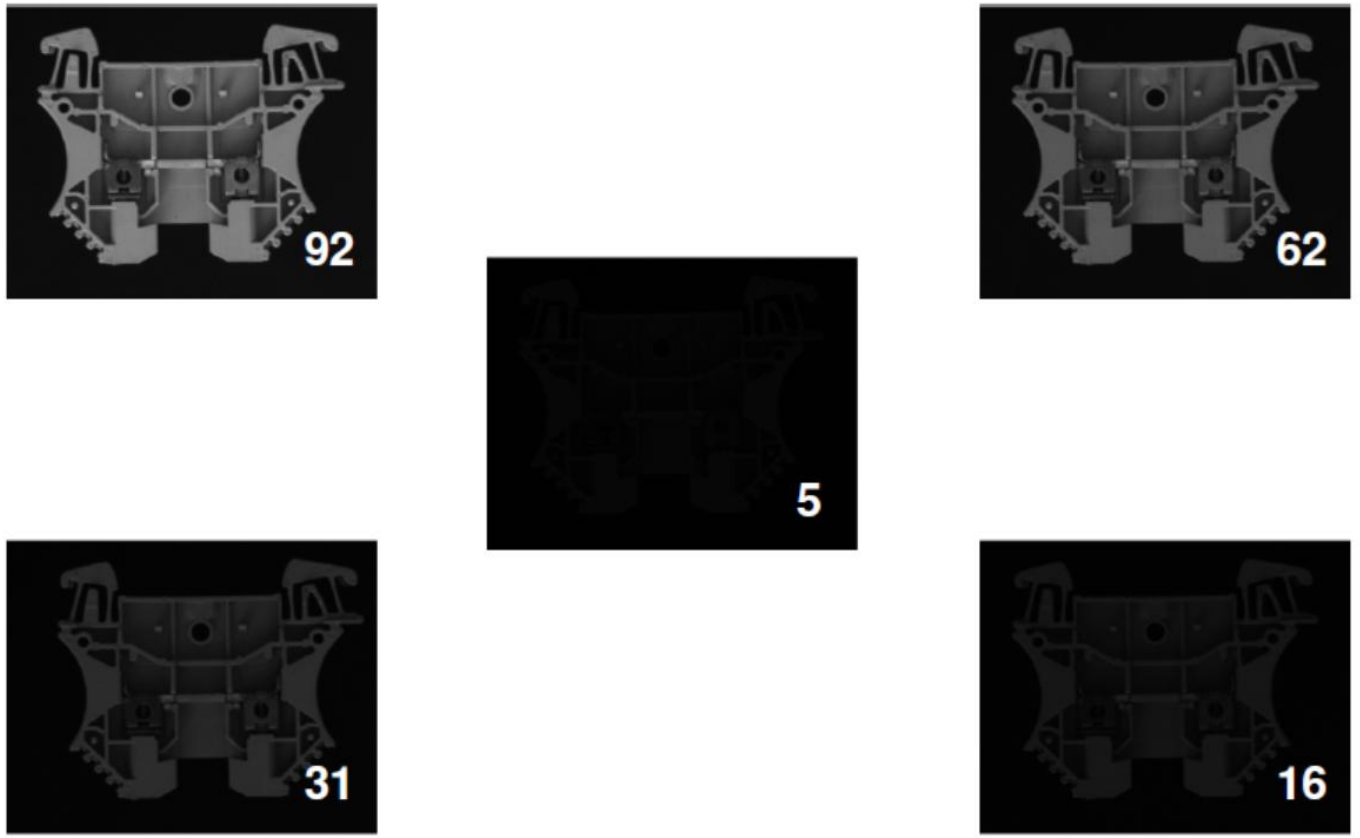
- **Accept Threshold** defines the **minimum degree of similarity** required between the **model pattern** and a **found pattern** for the pattern to be accepted.
- **Increasing the Accept Threshold can reduce the time required for the search** because fewer candidate patterns need to be evaluated or accepted.





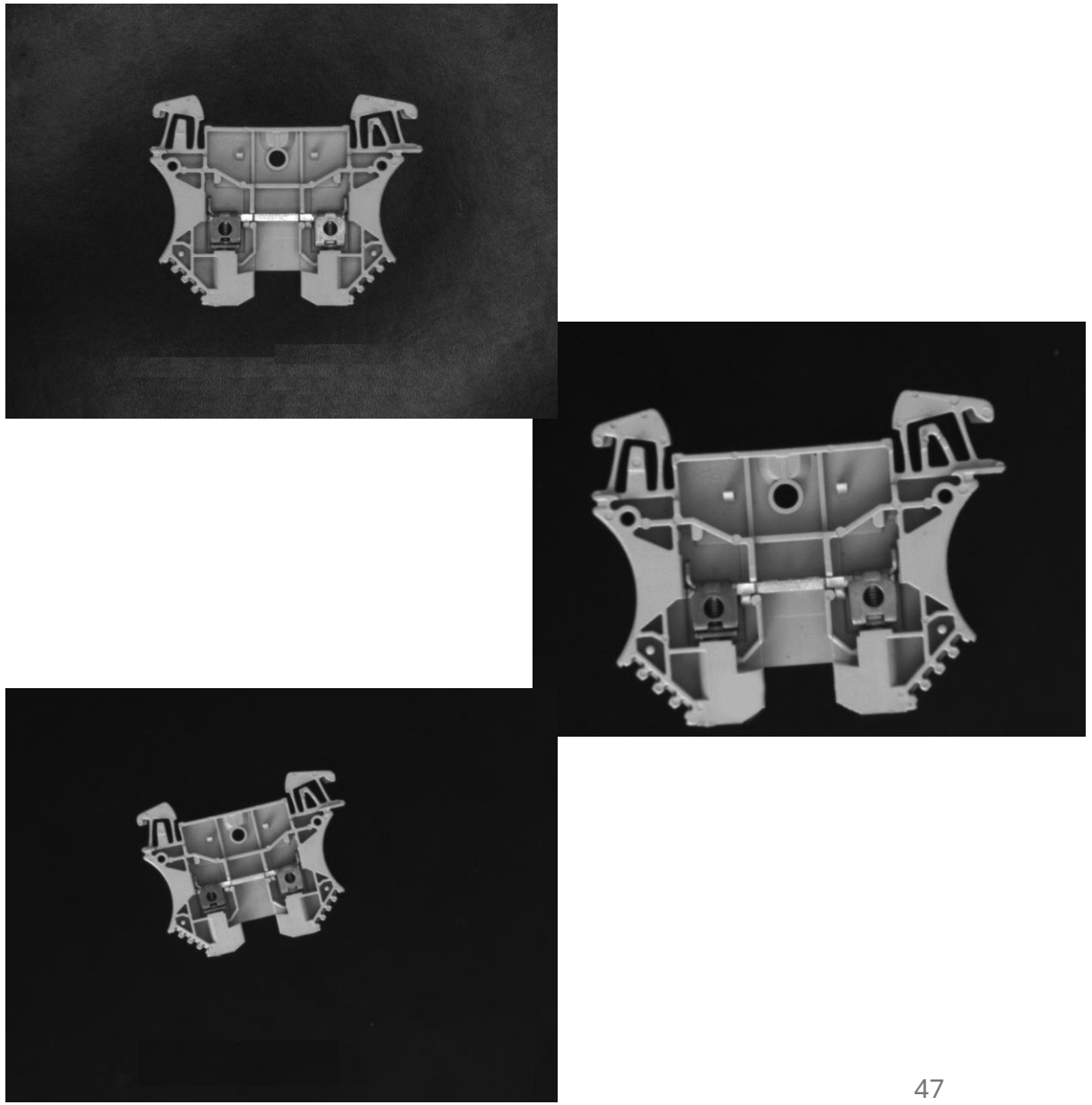
PatMax Pattern

- **Contrast Threshold** defines the **minimum acceptable contrast** that must be present in a found pattern.
- The contrast of a found pattern must be **greater than the Contrast Threshold value** for the pattern to be considered a **valid instance**.
- A **low Contrast Threshold** is used for **low-contrast images**, while a **high Contrast Threshold** is used for **high-contrast images**.



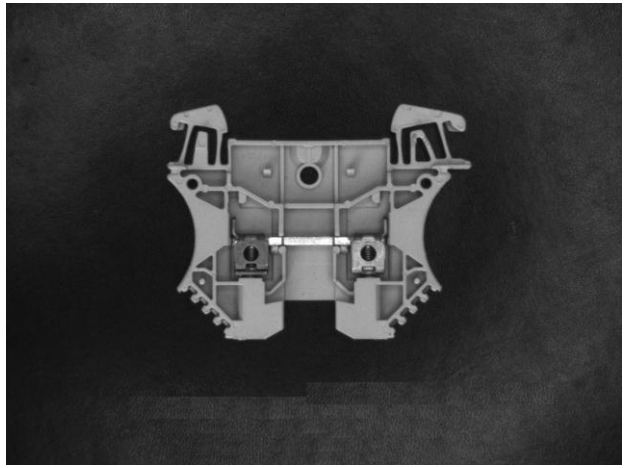
PatMax Pattern

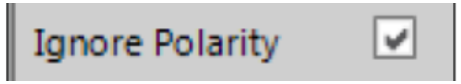
- **Scale Tolerance** defines the allowable **percentage of size variation** between the **found pattern** and the **model pattern** (**0–50%; default = 0%**).
- The value represents a **± percentage**. For example, a value of **5%** allows the tool to find patterns that are between **95% and 105%** of the *Model* pattern's size.
- Allowing a greater range of **scale variation** increases the tool's **execution time**.

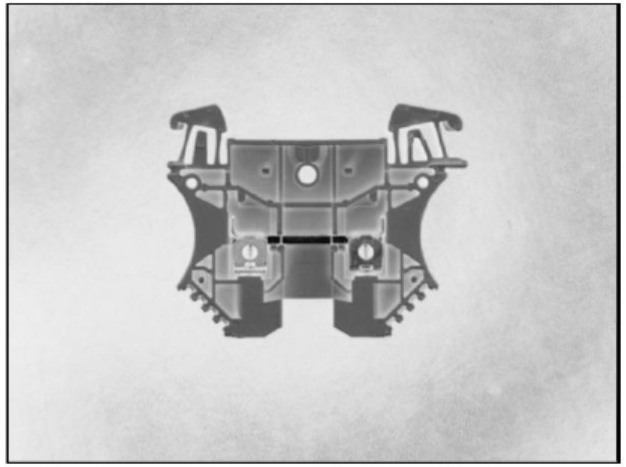


PatMax Pattern

- **Ignore Polarity** determines whether found patterns can contain **color-inverted features** relative to the **Model pattern** (default = **Off**, unchecked).
- When **Ignore Polarity** is enabled (**On**, checked), a pattern with **inverted feature polarity** (for example, black/white instead of white/black in the Model pattern) can still be **classified as a match** to the Model pattern.
- **Ignore Polarity** instructs PatMax to check for both the **original polarity identified during training** and the **opposite polarity**. This may **increase PatMax execution time**.

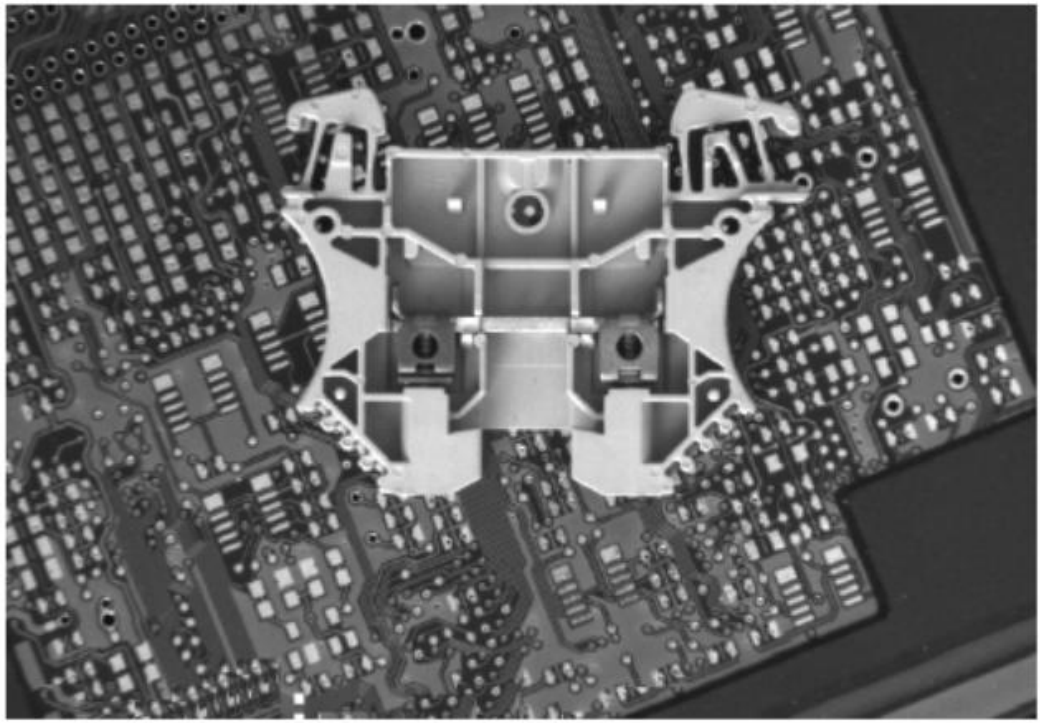






PatMax Pattern

- **Strict Scoring** determines whether **missing or occluded features** of the found pattern affect its score (default = **Off**, unchecked).
- When **Strict Scoring** is enabled (**On**, checked), missing or occluded features **lower the score** of the found pattern.

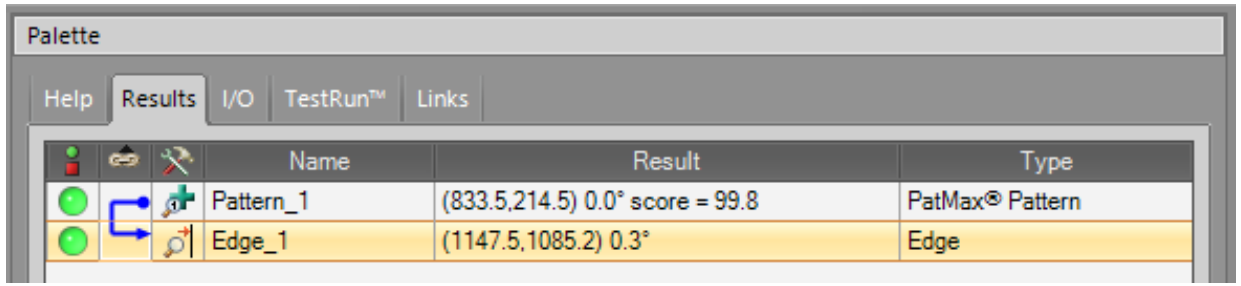




PatMax Pattern

- The image displays the **model features** and a **green cross** indicating the **center of the model**.
- The **Results table** displays information about each tool in the **active job**, allowing you to **monitor and troubleshoot the tools as the job runs**.
- The **PatMax Pattern tool** reports the results as **X, Y, Angle, and Score**.

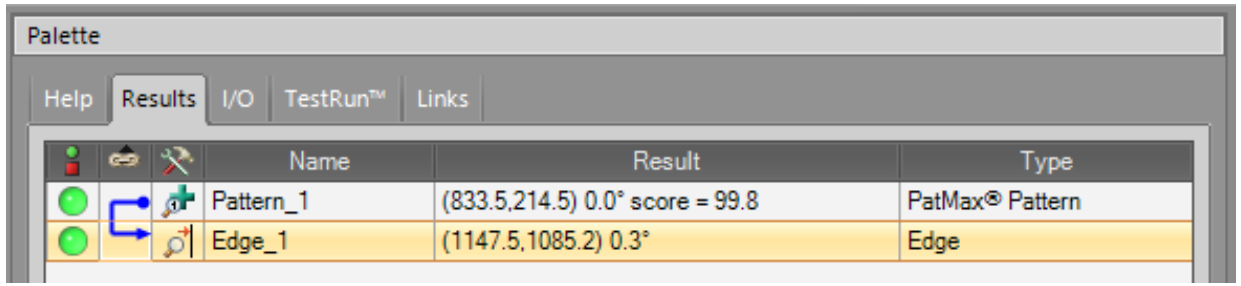




PatMax Pattern

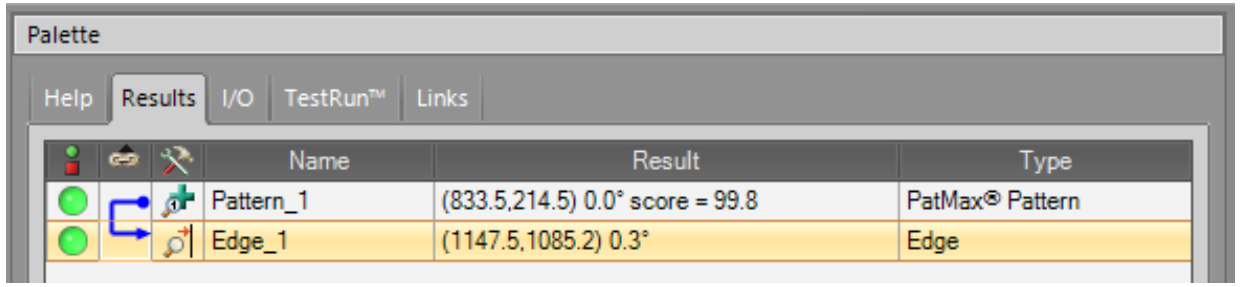
- For each tool, the **Results Table** displays its **pass/fail status, tool icon, name, result, pass total, fail total**, and the **time, in milliseconds, required to run the tool**.
- In addition, when a tool is selected in the table, the **dependencies** between the selected tool (highlighted in **orange**) and other tools in the job are displayed. This allows you to determine whether a **tool failure is related to the failure of another tool**. A **dependency** exists when one tool **references another tool**.

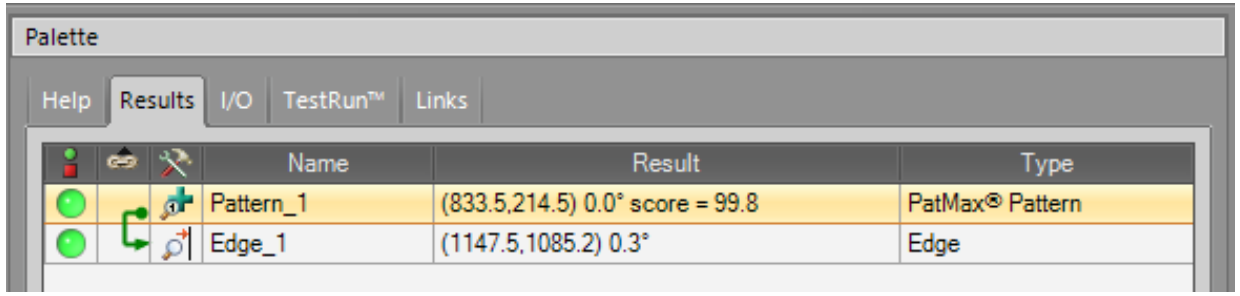




PatMax Pattern

- Each dependency consists of two parts: a **precedent** and a **dependent**, which are connected by a line.
- The **Precedent** is the **source of the data** (the tool providing the input) and is represented by a **circle**.
- The **Dependent** is the **destination of the data** (the tool receiving the input) and is represented by an **arrow**.
- **Blue dependency graphics** indicate that the selected tool is the **destination**, while **green dependency graphics** indicate that the selected tool is the **source**.





PatMax RedLine

- A complete reinvention of pattern matching. It is not based on PatMax.
- Available as an option on newer camera models.
- Generally 4–7 times faster than PatMax.
- Provides both high speed and high accuracy.

